ORBIT MVP walk-through

BMO Lead

Runs the board. Loads and gates the data, receives and accepts files, locks the population, and certifies the Order of Merit. Does not score.

Every screen this role can open, what each widget does, the data behind it, and how far each number can be trusted today.

What the BMO Lead can open

8 of 8 tools. In the live demo this role runs steps 1, 2, 6.

Tool	Access	What that access means	What the tool is for
HELIOS Command	**Owns**	Runs this page's actions.	Data in. Shows whether records loaded cleanly, how fresh each source is, and whether the board is ready to lock.
Candidate Queue	**View**	Read only.	The board's list of records. Shows how the list was built and where each record stands.
Panel Workspace	**View**	Read only.	One record, read through eight lenses, with the AI summary beside it and the scoring dock below.
Panel Oversight	**View**	Read only.	The Board President's controls: what the panel sees from AI and when, panel progress, re-votes and record issues.
Scoring Audit	**Uses**	Runs some actions here.	Finds scores far from the panel average and records what a person decided to do about each.
Board Statistics & Roles	**Owns**	Runs this page's actions.	One board's numbers, gates and role rights. This is where the population is locked and the Order of Merit is certified.
DELTA Ledger	**View**	Read only.	The audit trail of every action, and the gate every incoming file passes through.
Failure Tests	**View**	Read only.	One screen that answers: how does ORBIT test for failure, and is anything failing right now?

How to read these slides

Each slide is one part of a real screen, captured as this role.

On every slide

1 A numbered marker on the screenshot matches the numbered note beside it.

Shows	What the widget puts in front of the user.
Data	The object and fields it reads, and the source system in production.
Rule	The calculation or rule that turns the data into what you see.
Trust	How far the value can be relied on today. See the five words below.
BMO Lead	What this role can and cannot do with the widget.

The five trust words

- **Calculated live**
  Worked out from the records each time the screen draws. Change a record and the number changes.
- **Fixed rule**
  A yes or no rule or a set limit. Same input, same result, every time.
- **Logged to DELTA**
  The action writes an entry to the audit ledger.
- **Sample value**
  A number typed into the demo to show the layout. It is not calculated and will not change.
- **Pre-written stand-in**
  Fictional text or a number standing in for AI output. No model runs in the demo.

The frame around every screen: Header, tabs and footer (1 of 2)

The header, tabs and footer are the same on every page. They show who you are, what you may open, and whether the audit chain is intact.



1 Perspective menu (role)

Shows The role you are acting as. In the demo you pick it. In production it comes from your CAC sign-in and group.

Data Identity adapter: user ID to role table (5 roles in the MVP).

Rule A user with no role entry sees no tools. Every role change writes a ROLE_CHANGE entry.

Trust Fixed rule

2 Chain status badge

Shows Whether the DELTA ledger still verifies from first entry to last.

Data DeltaEntry.prevHash and hash for every entry.

Rule Recomputes each SHA-256 hash and compares it to the next entry's prevHash. Turns red on any break.

Trust Calculated live

The frame around every screen: Header, tabs and footer (2 of 2)

The header, tabs and footer are the same on every page. They show who you are, what you may open, and whether the audit chain is intact.



3 Catalog and Demo Guide

Shows Catalog lists the tools this role may open and the ones held for later. Demo Guide lists the walk-through steps for this role.

Data Tool registry and permission table.

Rule Both lists are filtered by the same permission table as the tabs.

Trust Fixed rule

4 Tool tabs

Shows One tab per tool this role may open, in the order a board moves.

Data Permission table (src/auth/permissions.ts).

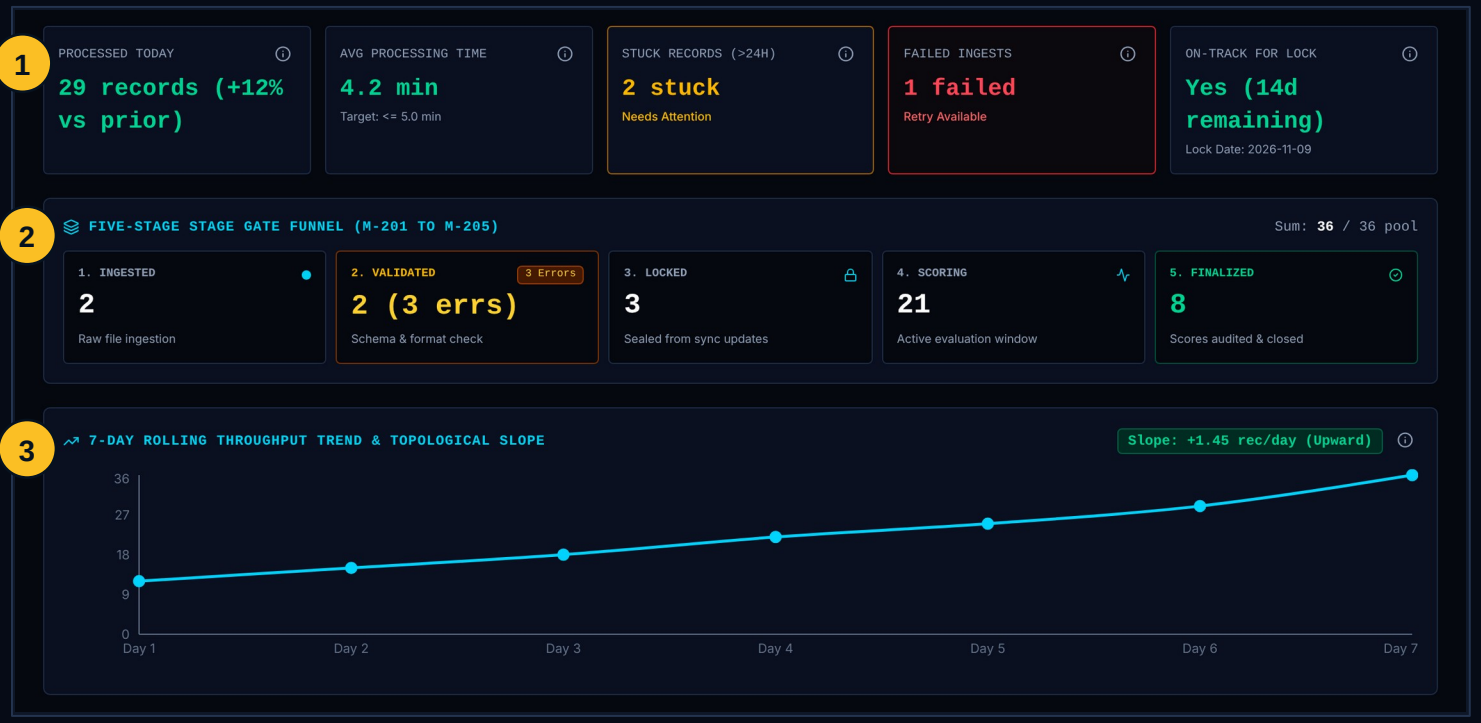
Rule Hiding a tab is a convenience. Each screen checks the role again when it opens and logs a refusal.

Trust Fixed rule

HELIOS Command: A. Pipeline and funnel (1 of 2)

Data in. Shows whether records loaded cleanly, how fresh each source is, and whether the board is ready to lock.

BMO Lead: Owns Runs this page's actions.



1 KPI strip (M-101 to M-105)

Shows Five headline numbers for the load pipeline: processed, average time, stuck, failed, on track for lock.

Data Candidate.flow: stage, errors, failedIngest.

Rule Processed = records in Scoring or Finalized. Stuck = Ingested with no failure, or Validated with errors. Failed = failedIngest is true.

Trust Calculated live for the processed count, stuck and failed. Sample value for "+12% vs prior", "4.2 min" and "Yes (14d remaining)".

2 Five-stage funnel (M-201 to M-205)

Shows How many records sit at each stage: Ingested, Validated, Locked, Scoring, Finalized.

Data Candidate.flow.stage, Candidate.flow.errors.

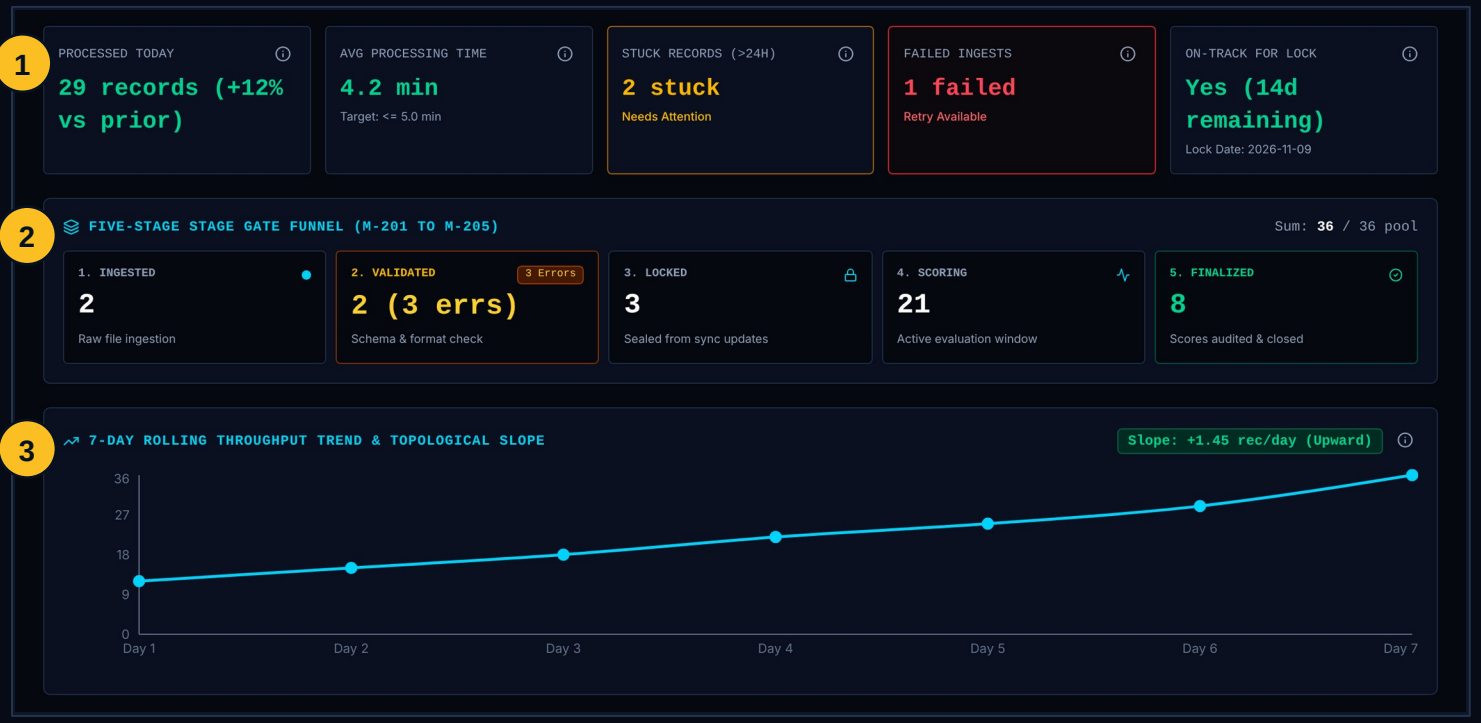
Rule Count of records per stage. The Validated card also counts schema errors. The five counts must add up to the pool (36).

Trust Calculated live

HELIOS Command: A. Pipeline and funnel (2 of 2)

Data in. Shows whether records loaded cleanly, how fresh each source is, and whether the board is ready to lock.

BMO Lead: Owns Runs this page's actions.



3 Throughput trend (M-106)

Shows Records processed per day over seven days, with the trend slope.

Data None live in the demo.

Rule Production rule: least-squares slope of the 7-day processed counts. The slope function exists and is unit tested.

Trust Sample value. The seven points and the +1.45 slope are fixed in the demo.

HELIOS Command: A. Stuck and failed records

Data in. Shows whether records loaded cleanly, how fresh each source is, and whether the board is ready to lock.

BMO Lead: Owns Runs this page's actions.



1 Stuck records list

Shows Each record that has not moved, with its stage and the reason. Open jumps to the record.

Data Candidate.flow.stage and flow.errors.

Rule Same rule as M-103. The demo does not measure elapsed hours; it uses stage plus errors.

Trust Calculated live

BMO Lead Opens the record to work the error.

2 Failed ingests list

Shows Records whose load failed, with the error text, and a Retry button.

Data Candidate.flow.failedIngest and flow.errors.

Rule Retry clears the failure, moves the record to Validated and writes PIPELINE_RETRY to DELTA.

Trust Calculated live + logged to DELTA

BMO Lead Runs Retry.

HELIOS Command: B. Source freshness

Data in. Shows whether records loaded cleanly, how fresh each source is, and whether the board is ready to lock.

BMO Lead: Owns Runs this page's actions.



1 Source freshness monitor (M-301 to M-304)

Shows For ARMS, MilPDS, MyVector and SURF: time since last sync, a bar against the limits, and a status. Selecting a source lists the records it affects.

Data SourcePipeline: last sync time, amber and red limits.

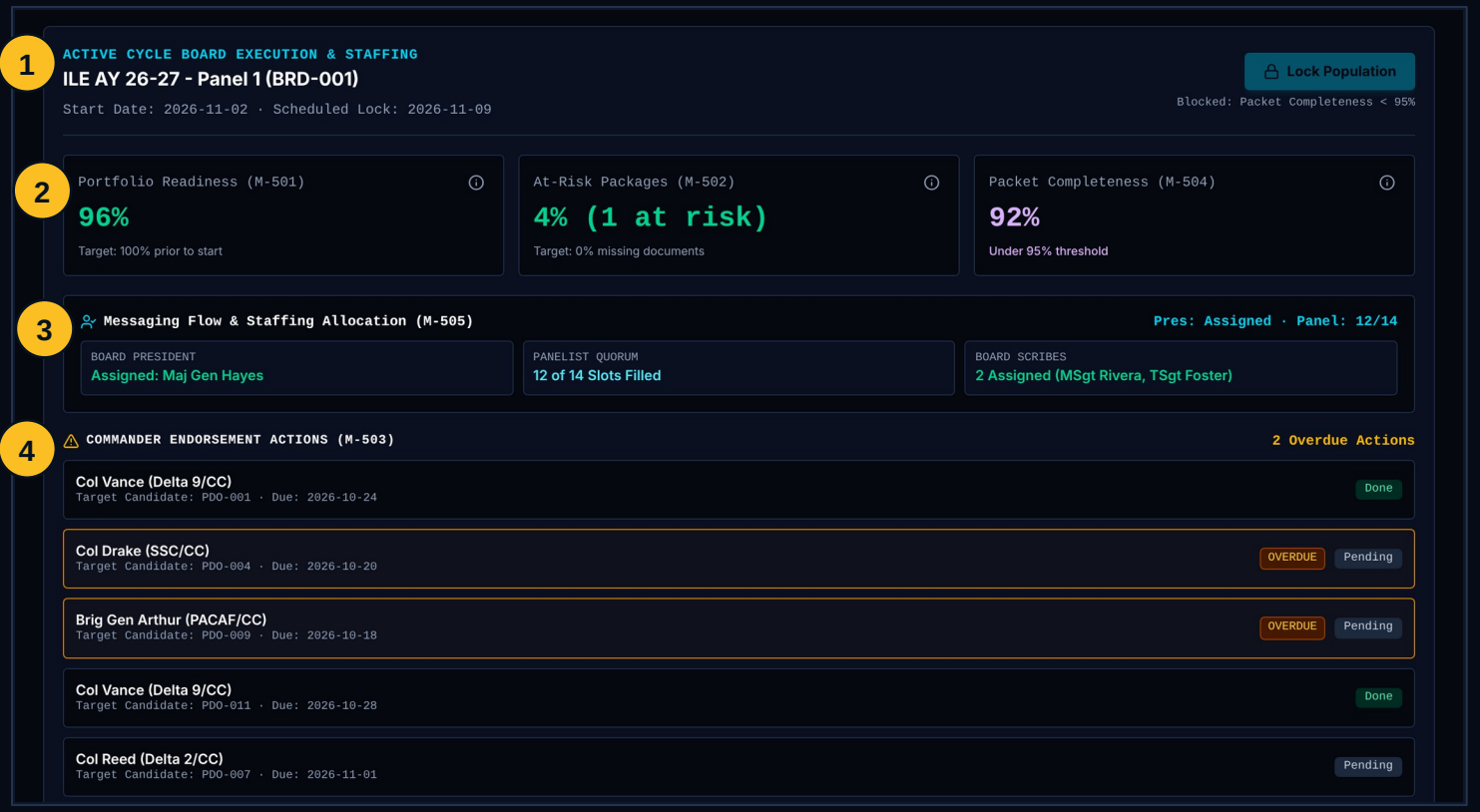
Rule Limits: ARMS amber over 12h, red over 24h. MilPDS 24h and 48h. MyVector 24h and 72h. SURF 1 day and 7 days.

Trust Fixed rule for the limits. Calculated live for the ages: last sync to the data snapshot time (5.2h, 32.7h, 32 min, 8.5 days). The sync times are fictional.

HELIOS Command: C. Readiness and lock (1 of 2)

Data in. Shows whether records loaded cleanly, how fresh each source is, and whether the board is ready to lock.

BMO Lead: Owns Runs this page's actions.



1 Board card and Lock Population

Shows The board, its start and lock dates, and the Lock Population button.

Data Board; metric M-504.

Rule The button works only for the BMO Lead and only when packet completeness is 95% or more. Locking writes POPULATION_LOCK.

Trust Fixed rule + logged to DELTA

BMO Lead Locks the population once the 95% gate is met.

2 Readiness gauges (M-501, M-502, M-504)

Shows Portfolio readiness, at-risk packages, and packet completeness against the 95% lock gate.

Data Candidate.packetReadiness.complete.

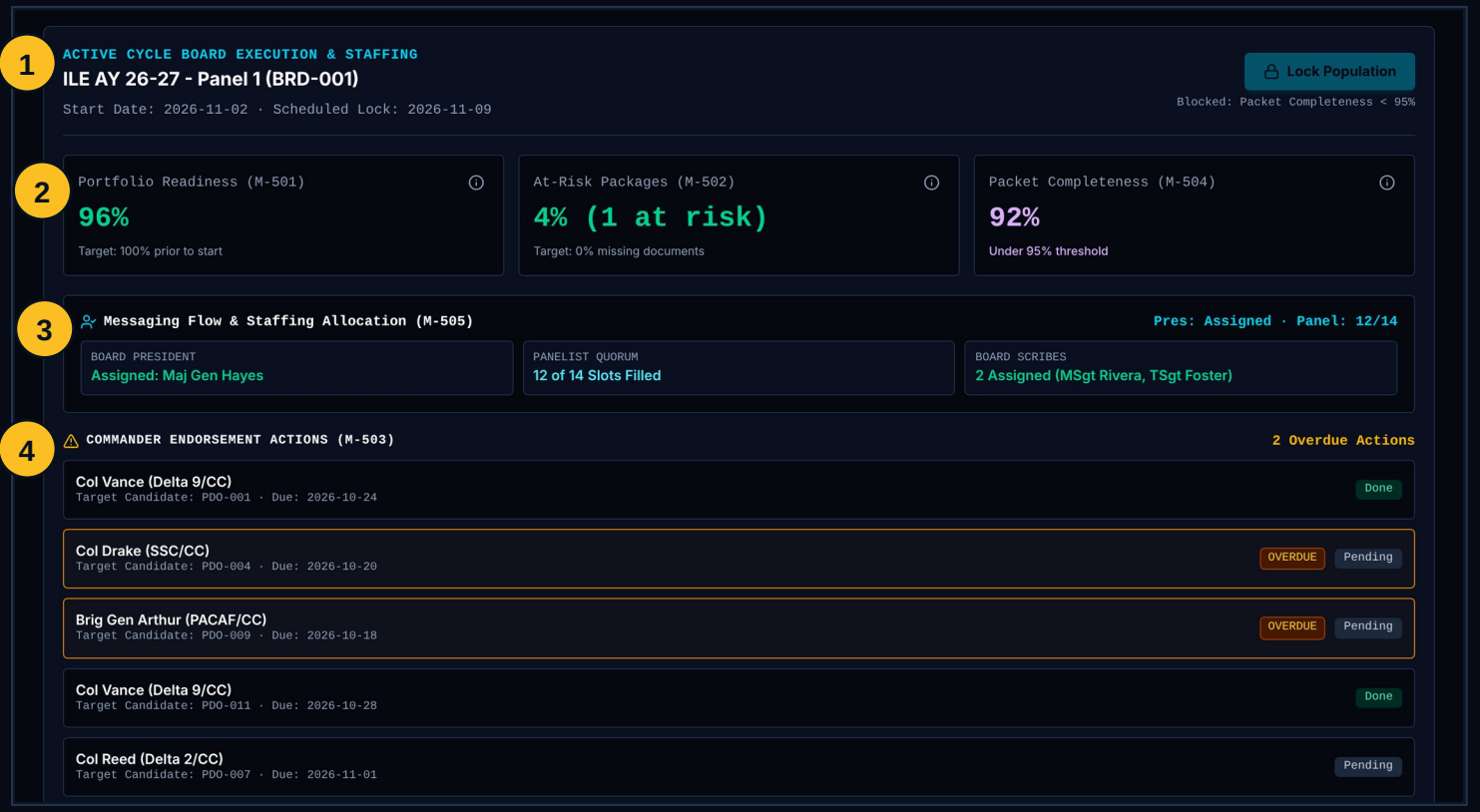
Rule Packet completeness = complete packets divided by records (33 of 36 = 92%).

Trust Calculated live for packet completeness. Sample value for readiness 96% and at-risk 4%.

HELIOS Command: C. Readiness and lock (2 of 2)

Data in. Shows whether records loaded cleanly, how fresh each source is, and whether the board is ready to lock.

BMO Lead: Owns Runs this page's actions.



3 Staffing (M-505)

Shows Whether a President is assigned, how many panelist seats are filled, and the scribes.

Data StaffSlot: role, assigned, name.

Rule Counts assigned seats by role. Green at 12 or more panelists.

Trust Calculated live

4 Commander actions (M-503)

Shows Endorsement actions owed by commanders, with due dates and overdue flags.

Data CommanderAction: record, commander, due, status.

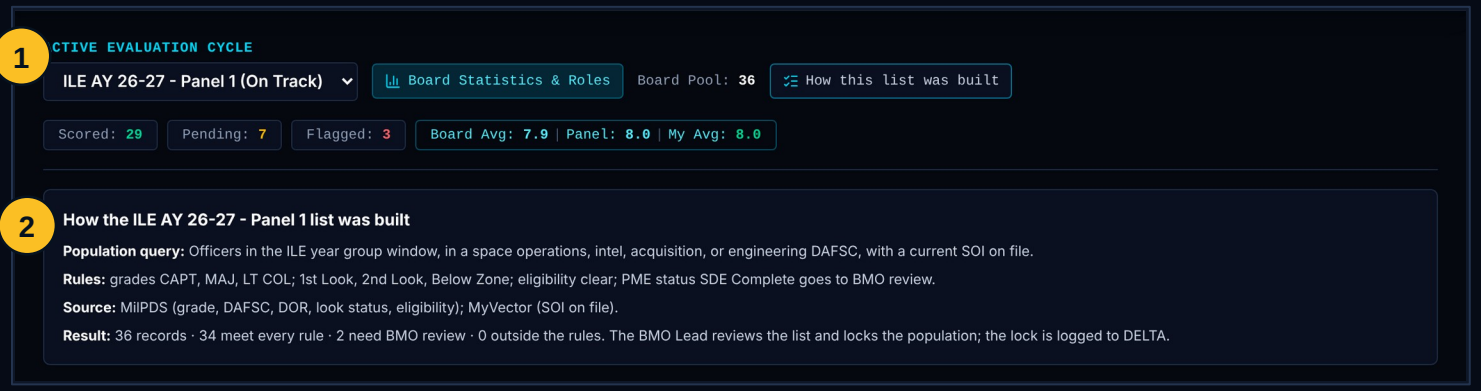
Rule Overdue = Pending and due before the demo's reference date (26 Oct 2026).

Trust Calculated live against a fixed reference date

Candidate Queue: Header, counters and roster rules (1 of 2)

The board's list of records. Shows how the list was built and where each record stands.

BMO Lead: View Read only.



1 Board header and counters

Shows The board, pool size, and counts of Scored, Pending and Flagged for the signed-in voter, plus three averages.

Data Candidate.scores, Candidate.myScore, Candidate.flagged.

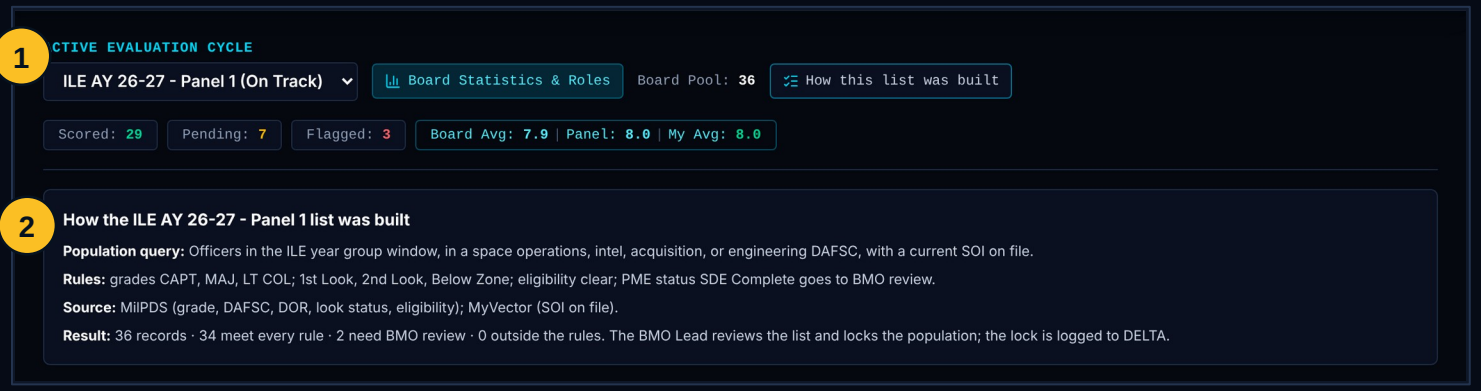
Rule Board Avg = mean of every score on the board. My Avg = mean of your own scores.

Trust Calculated live for counts, Board Avg and My Avg. Sample value for "Panel": it is Board Avg plus 0.1, and My Avg shows a placeholder until you score.

Candidate Queue: Header, counters and roster rules (2 of 2)

The board's list of records. Shows how the list was built and where each record stands.

BMO Lead: View Read only.



2 How this list was built

Shows The roster rule for this board and how many records were Included, sent to BMO review, or Excluded.

Data Roster rule (grades, look status, PME review); Candidate grade, lookStatus, eligibility, pmeStatus.

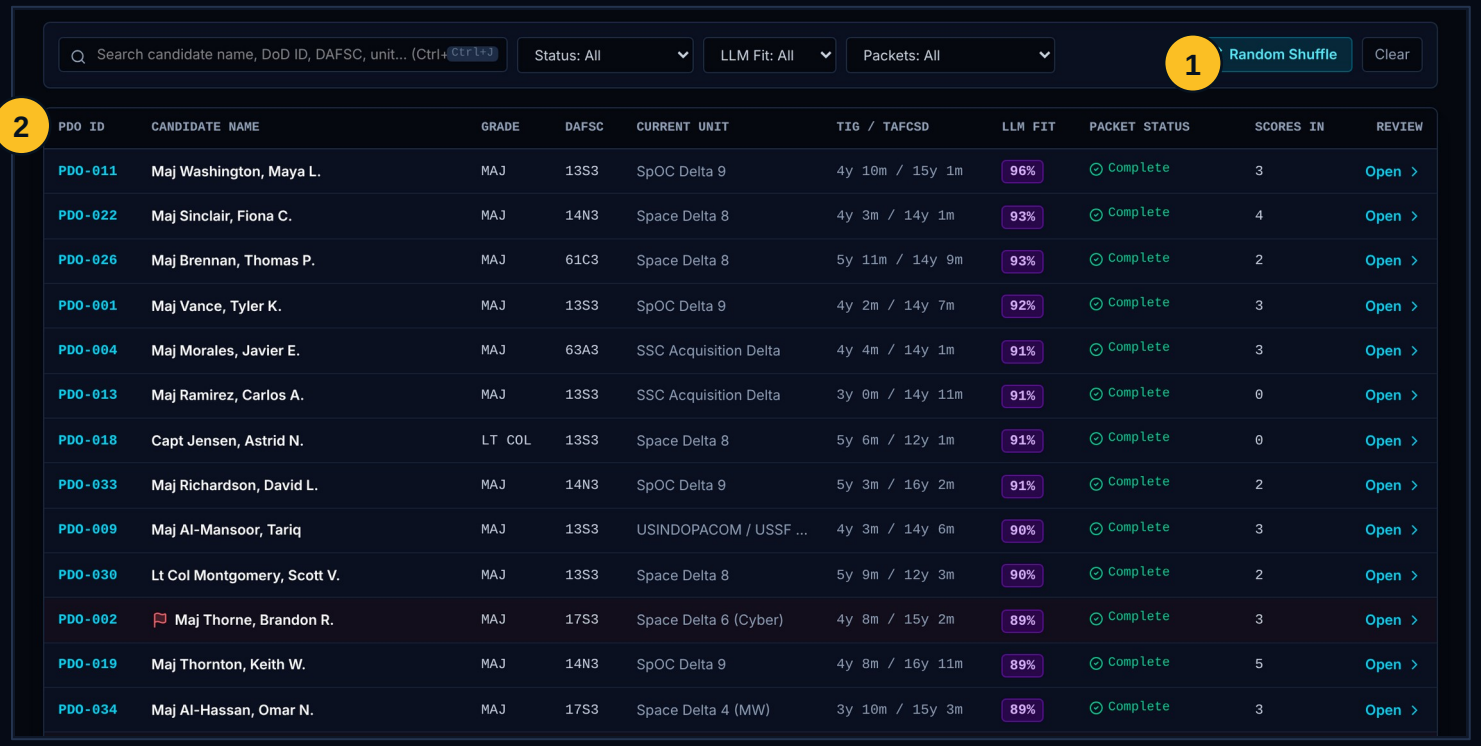
Rule Outside the grade or look rule = Excluded. Eligibility issue or listed PME status = BMO review. Otherwise Included. Production source: MilPDS query plus SOI on file in MyVector.

Trust Calculated live (same function runs in the unit tests)

Candidate Queue: Filters, shuffle and the record table (1 of 2)

The board's list of records. Shows how the list was built and where each record stands.

BMO Lead: View Read only.



PDO ID	CANDIDATE NAME	GRADE	DAFSC	CURRENT UNIT	TIG / TAFCSD	LLM FIT	PACKET STATUS	SCORES IN	REVIEW
PDO-011	Maj Washington, Maya L.	MAJ	13S3	SpOC Delta 9	4y 10m / 15y 1m	96%	Complete	3	Open
PDO-022	Maj Sinclair, Fiona C.	MAJ	14N3	Space Delta 8	4y 3m / 14y 1m	93%	Complete	4	Open
PDO-026	Maj Brennan, Thomas P.	MAJ	61C3	Space Delta 8	5y 11m / 14y 9m	93%	Complete	2	Open
PDO-001	Maj Vance, Tyler K.	MAJ	13S3	SpOC Delta 9	4y 2m / 14y 7m	92%	Complete	3	Open
PDO-004	Maj Morales, Javier E.	MAJ	63A3	SSC Acquisition Delta	4y 4m / 14y 1m	91%	Complete	3	Open
PDO-013	Maj Ramirez, Carlos A.	MAJ	13S3	SSC Acquisition Delta	3y 0m / 14y 11m	91%	Complete	0	Open
PDO-018	Capt Jensen, Astrid N.	LT COL	13S3	Space Delta 8	5y 6m / 12y 1m	91%	Complete	0	Open
PDO-033	Maj Richardson, David L.	MAJ	14N3	SpOC Delta 9	5y 3m / 16y 2m	91%	Complete	2	Open
PDO-009	Maj Al-Mansoor, Tariq	MAJ	13S3	USINDOPACOM / USSF ...	4y 3m / 14y 6m	90%	Complete	3	Open
PDO-030	Lt Col Montgomery, Scott V.	MAJ	13S3	Space Delta 8	5y 9m / 12y 3m	90%	Complete	2	Open
PDO-002	Maj Thorne, Brandon R.	MAJ	17S3	Space Delta 6 (Cyber)	4y 8m / 15y 2m	89%	Complete	3	Open
PDO-019	Maj Thornton, Keith W.	MAJ	14N3	SpOC Delta 9	4y 8m / 16y 11m	89%	Complete	5	Open
PDO-034	Maj Al-Hassan, Omar N.	MAJ	17S3	Space Delta 4 (MW)	3y 10m / 15y 3m	89%	Complete	3	Open

1 Search, filters and Random Shuffle

Shows Search by name, ID, DAFSC or unit. Filter by status and packet check. Random Shuffle reorders the list.

Data Candidate fields; browser random source.

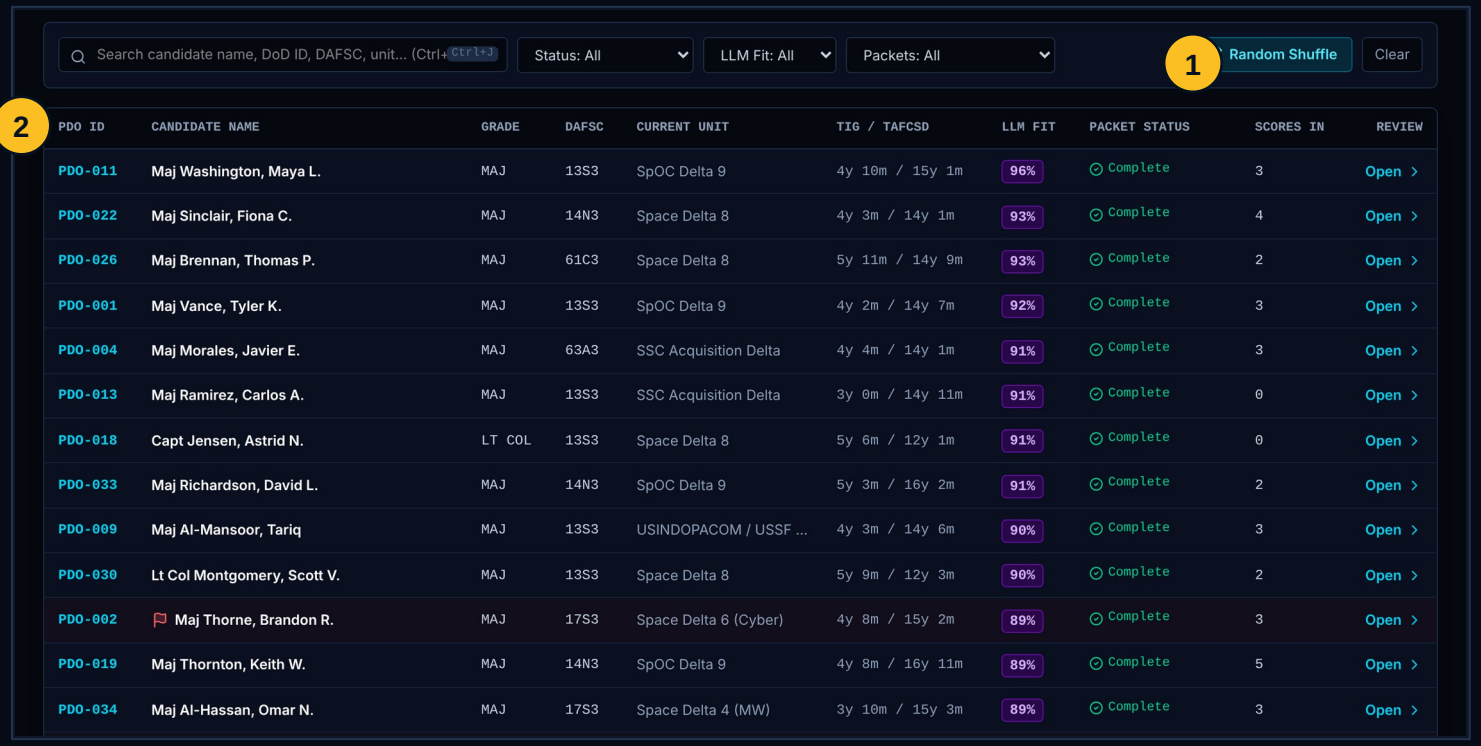
Rule Shuffle is a Fisher-Yates reorder seeded from the browser's cryptographic random source. The seed is written to DELTA as SHUFFLE so the order can be reproduced.

Trust Fixed rule + logged to DELTA

Candidate Queue: Filters, shuffle and the record table (2 of 2)

The board's list of records. Shows how the list was built and where each record stands.

BMO Lead: View Read only.



2 Record table

Shows One row per record: ID, name, grade, DAFSC, unit, time in grade and service, AI fit, packet status, your score. A row opens the record in Panel Workspace.

Data Candidate; roster check; Candidate.packetReadiness.

Rule Packet status lists exceptions. "BMO review" comes from the roster rule. AI fit follows the President's visibility setting.

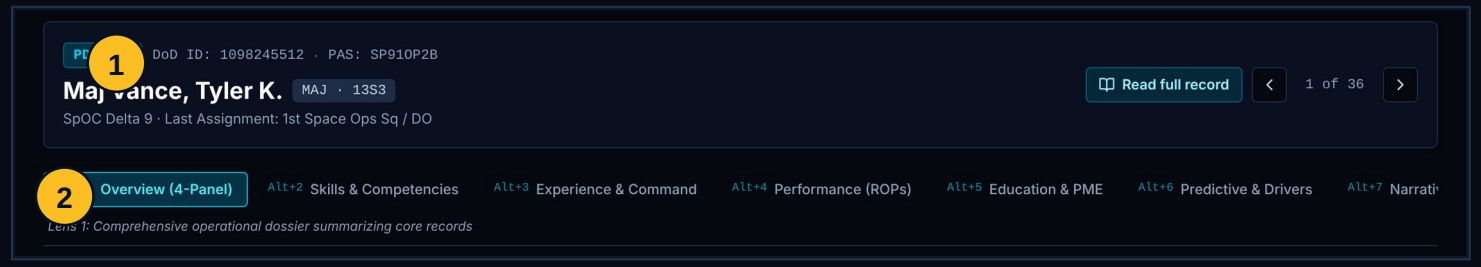
Trust Calculated live for packet status and scores. Pre-written stand-in for the AI fit value: a number typed per record, not produced by a model.

BMO Lead Sees AI fit at all times. Does not score.

Panel Workspace: Record header and the eight lenses

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 Record header

Shows Record ID, name, grade and DAFSC, unit, last assignment. Buttons: Read full record, Discuss, previous and next.

Data Candidate identity fields. Full text: SOI, commander comments, appraisals.

Rule Opening a full-text document writes RECORD_OPEN. Chat posts are blocked if they contain health-related words.

Trust Calculated live + logged to DELTA

BMO Lead Reads the record. Does not see panel chat. Cannot score.

2 Lens tabs (Alt+1 to Alt+8)

Shows Eight views of the same record: Overview, Skills, Experience, Performance, Education, Predictive, Narrative and SOI, Eligibility and Freshness.

Data One Candidate object. A lens changes the view, never the data.

Rule Every voter sees the same eight lenses in the same order.

Trust Fixed rule

Panel Workspace: Lens 1: Overview (1 of 2)

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 Statement of Intent

Shows Opening lines of the SOI, its date, and topic tags.

Data Candidate.soi: text, date, tags. Production source: MyVector.

Rule Shown as written. Tags are stored with the record.

Trust Calculated live display of Pre-written stand-in fictional text

2 Evaluations and stratifications

Shows Last three years: stratification statement and percentile, with a trend label.

Data Candidate.performance, Candidate.ropTrend. Production source: evaluation system (to confirm).

Rule Shown as stored. The trend label is a stored field, not computed.

Trust Calculated live display; trend is Pre-written stand-in

3 ARMS and source freshness

Shows Badge level and the last sync time of the ARMS and SURF data for this record.

Data Candidate.badgeLevel, Candidate.freshness.

Rule Shown as stored, so the voter can see how old the data is.

Trust Calculated live display

Panel Workspace: Lens 1: Overview (2 of 2)

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



4 SURF career record

Shows Time in grade, time in service, deployments, joint and short tours, PME status.

Data Candidate.tig, tafcsd, deployments, jointTours, pmeStatus. Production source: MilPDS and SURF.

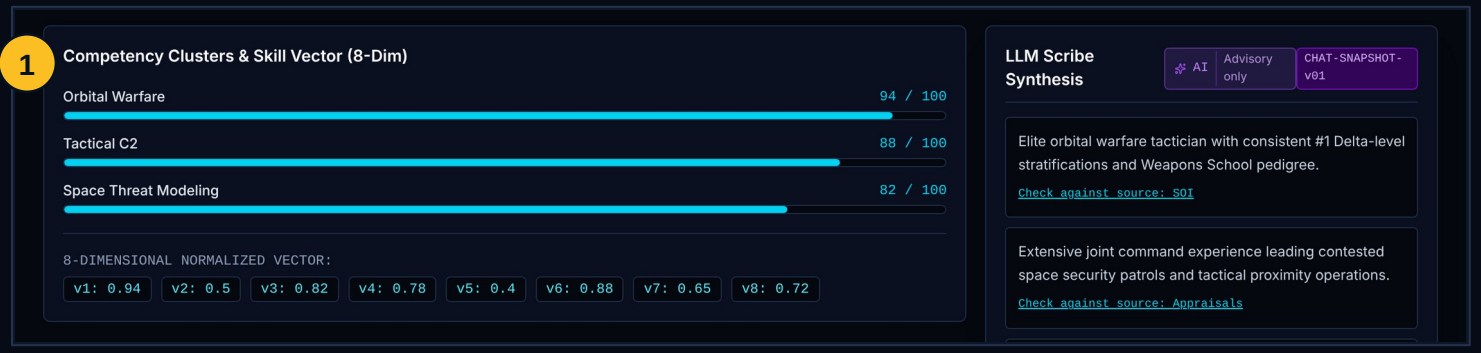
Rule Shown as stored.

Trust Calculated live display

Panel Workspace: Lens 2: Skills

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 Competency clusters and skill vector

Shows Skill clusters with a level each, and an eight-number skill vector for the record.

Data Candidate.skills, Candidate.skillVector (8 values from 0 to 1).

Rule The vector feeds the similar-record and billet-fit cards.

Trust Pre-written stand-in. The eight values are typed per record. No method yet derives them from a real record.

Panel Workspace: Lens 3: Experience

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 Command and duty chronology

Shows Duty history in date order: position, unit and dates.

Data Candidate.timeline. Production source: MilPDS duty history.

Rule Shown as stored, oldest to newest.

Trust Calculated live display

Panel Workspace: Lens 4: Performance

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 Performance reports, multi-year

Shows Each year's report: stratification, percentile and rating.

Data Candidate.performance.

Rule Shown as stored. No score is computed from it.

Trust Calculated live display

Panel Workspace: Lens 5: Education

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 Education, credentials and certifications

Shows Highest degree and discipline, PME status and year, badge, certifications.

Data Candidate.highestDegree, academicDiscipline, pmeStatus, pmeYear, certs. Production source: MilPDS education and training.

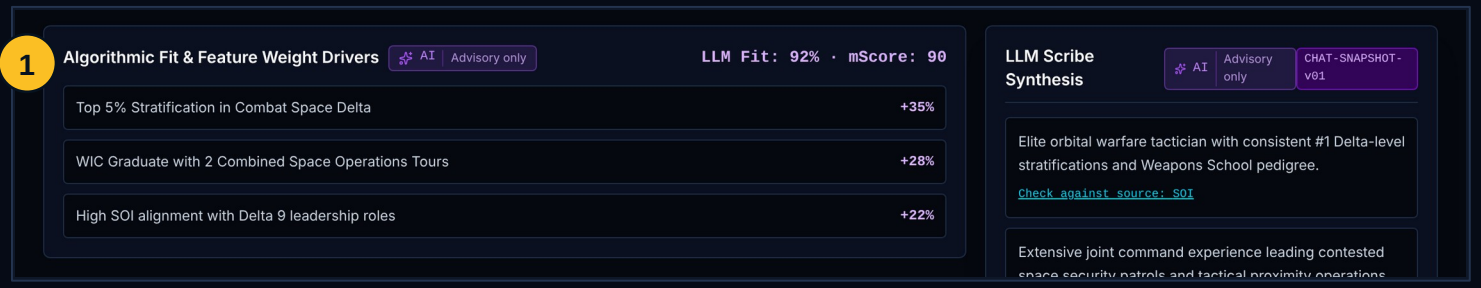
Rule Shown as stored.

Trust Calculated live display

Panel Workspace: Lens 6: Predictive and drivers

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 AI fit, model score and feature drivers

Shows An AI fit value, a model score, a confidence level per source, and the features that drove the score with their weights.

Data Candidate.llmFitScore, mScore, confidence, featureDrivers.

Rule Visibility follows the President's setting. Default: shown only after the voter scores the record.

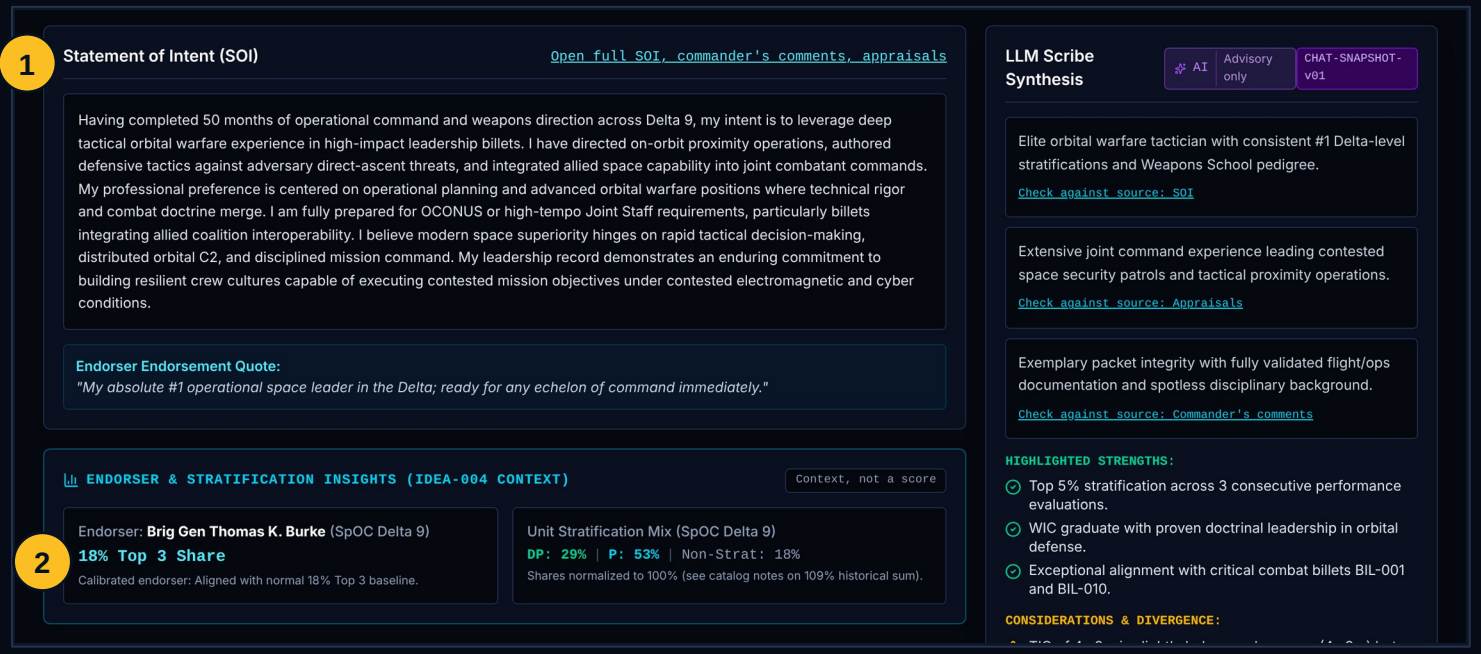
Trust Pre-written stand-in. All four are typed per record. No model runs in the demo, so these numbers have no validity evidence.

BMO Lead Always visible. Does not score.

Panel Workspace: Lens 7: Narrative and endorser

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 Statement of Intent, full

Shows The full SOI text, word count, tags, endorser quote and stated preferences.

Data Candidate.soi.

Rule Shown as written.

Trust Calculated live display of Pre-written stand-in fictional text

2 Endorser context (M-404, M-405)

Shows How often this endorser places people in their top 3, and the unit's mix of DP, P and non-stratified ratings.

Data Endorser.top3SharePct, Endorser.stratMix.

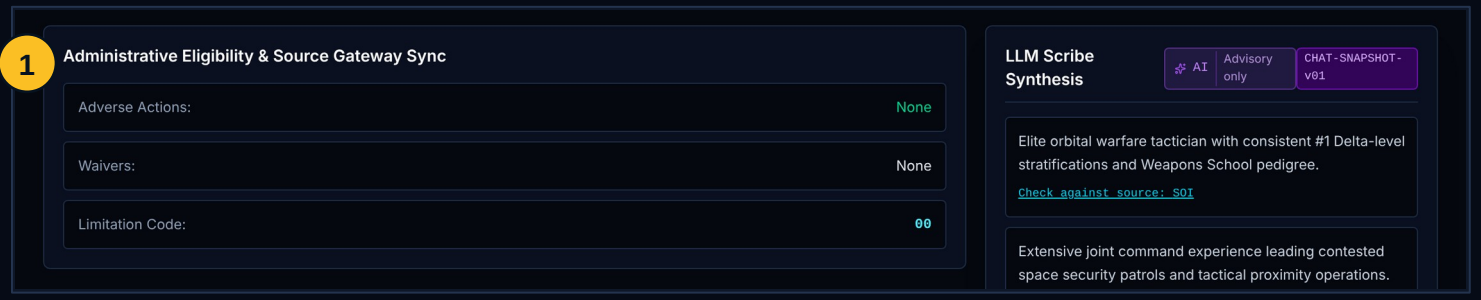
Rule Context for reading an endorsement. It does not change any score.

Trust Pre-written stand-in. Values are typed per endorser. A real version needs endorsement history from the source system.

Panel Workspace: Lens 8: Eligibility and freshness

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 Eligibility and source sync

Shows Adverse actions, waivers, limitation code, and the last sync time of each source for this record.

Data Candidate.eligibility, Candidate.freshness.

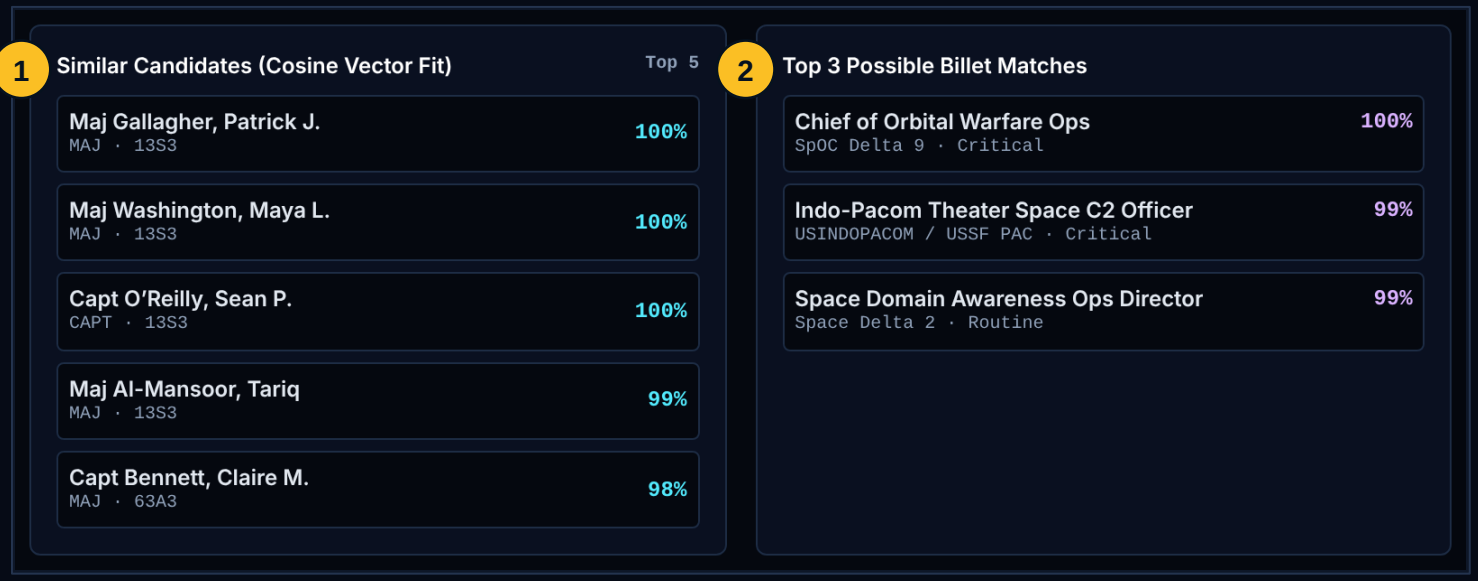
Rule Shown as stored. An eligibility issue also sends the record to BMO review in the roster rule.

Trust Calculated live display

Panel Workspace: Similar records and billet fits

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 Similar records (top 5)

Shows The five records most like this one, with a similarity percentage.

Data Candidate.skillVector for every record.

Rule Cosine similarity between two 8-value skill vectors; top 5 shown. The function is unit tested.

Trust Calculated live math on Pre-written stand-in vectors. Because all values are positive, scores bunch near 98 to 100% and separate records poorly.

2 Possible billet fits (top 3)

Shows The three positions whose skill vector is closest to this record's.

Data Candidate.skillVector, Billet.skillVector.

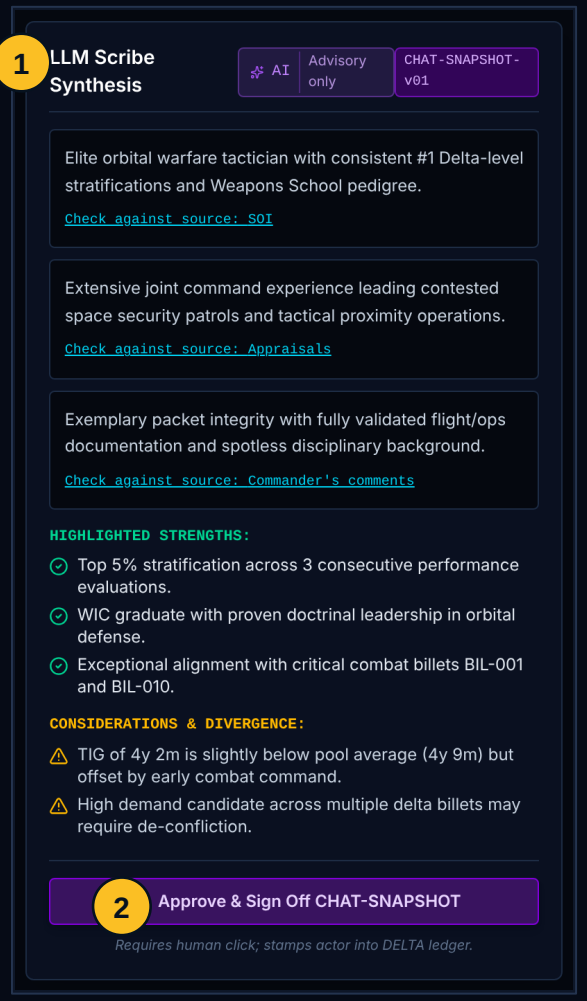
Rule Cosine similarity; top 3 shown. Advisory only. It never feeds the score or the Order of Merit.

Trust Calculated live math on Pre-written stand-in vectors

Panel Workspace: AI summary and human sign-off

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 AI summary (frozen copy)

Shows A three-paragraph summary, highlighted strengths, and considerations, each with a "check against source" link.

Data Candidate.chatSnapshot: summary, strengths, considerations.

Rule Marked "AI, advisory only". Each paragraph links to the source document so the reader can check it.

Trust Pre-written stand-in. The text is pre-written per record. No language model runs in the demo.

2 Human sign-off

Shows The button that records a named person approving the summary, or the name and time once signed.

Data chatSnapshot.signedOffBy, signedOffAt.

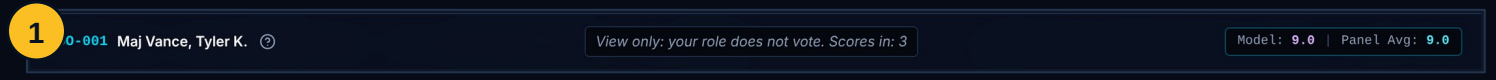
Rule Sign-off writes SNAPSHOT_SIGNOFF with the actor. It can be revoked, which writes SNAPSHOT_REVOKE. The Admin is refused.

Trust Fixed rule + logged to DELTA

Panel Workspace: Scoring dock

One record, read through eight lenses, with the AI summary beside it and the scoring dock below.

BMO Lead: View Read only.



1 Scoring dock

Shows Score buttons from 6.0 to 10.0 in half points, Submit and Next, Flag for Discussion, and a reveal area for the model score and panel average.

Data Candidate.scores (panelist ID, value, time).

Rule One original score per voter per record. A change needs a written reason and is logged as SCORE_REVISE. The model score and panel average stay hidden until you submit, to avoid anchoring.

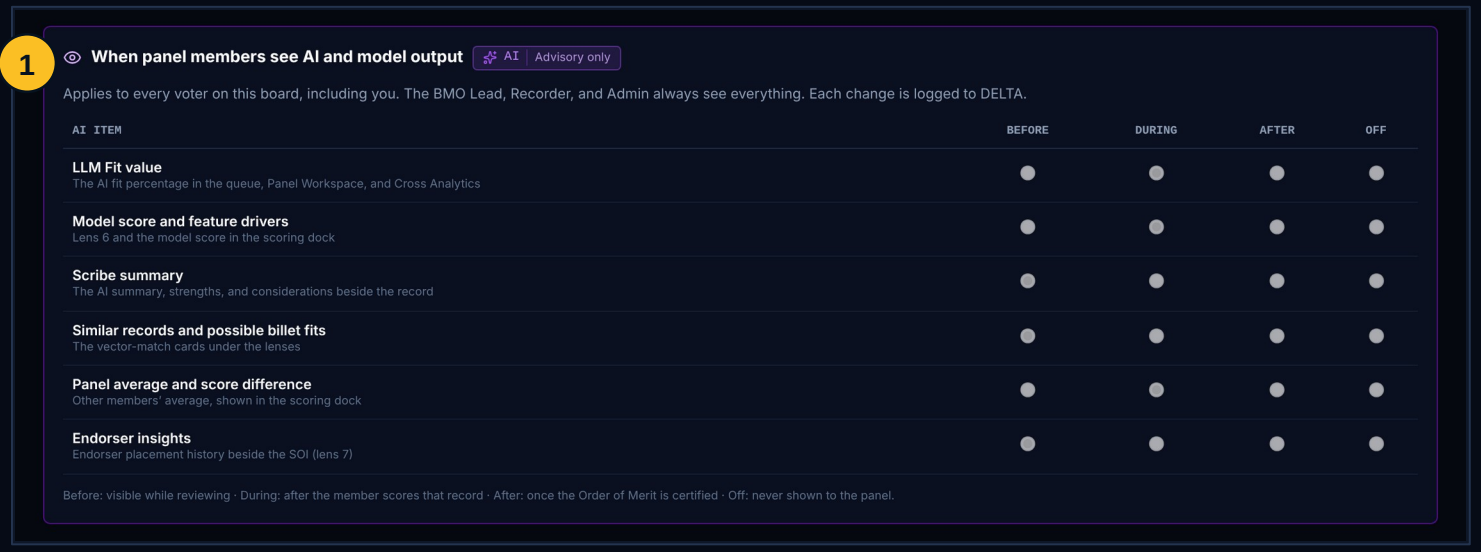
Trust Fixed rule + logged to DELTA (SCORE_SUBMIT)

BMO Lead No score buttons. Read only.

Panel Oversight: When panel members see AI output

The Board President's controls: what the panel sees from AI and when, panel progress, re-votes and record issues.

BMO Lead: View Read only.



1 AI visibility settings

Shows Six AI items, each set to Before, During, After or Off: fit value, model score and drivers, summary, similar records and billet fits, panel average, endorser insights.

Data AiVisibility per board.

Rule Before = while reviewing. During = after the member scores that record. After = once the Order of Merit is certified. Off = never. Saving writes AI_VISIBILITY_CHANGE.

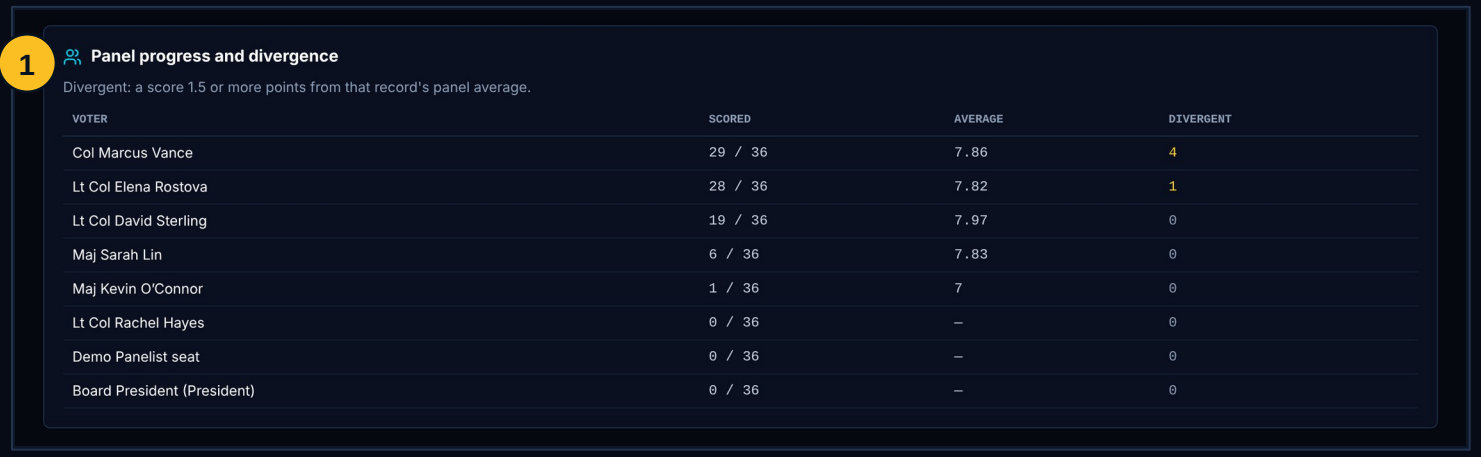
Trust Fixed rule + logged to DELTA. Demo default hides fit, model score and panel average until the member scores.

BMO Lead Reads the settings. Cannot change them.

Panel Oversight: Panel progress and divergence

The Board President's controls: what the panel sees from AI and when, panel progress, re-votes and record issues.

BMO Lead: View Read only.



1 Panel progress and divergence

Shows Per voter: records scored out of the pool, average score, and count of divergent scores.

Data Candidate.scores grouped by panelist.

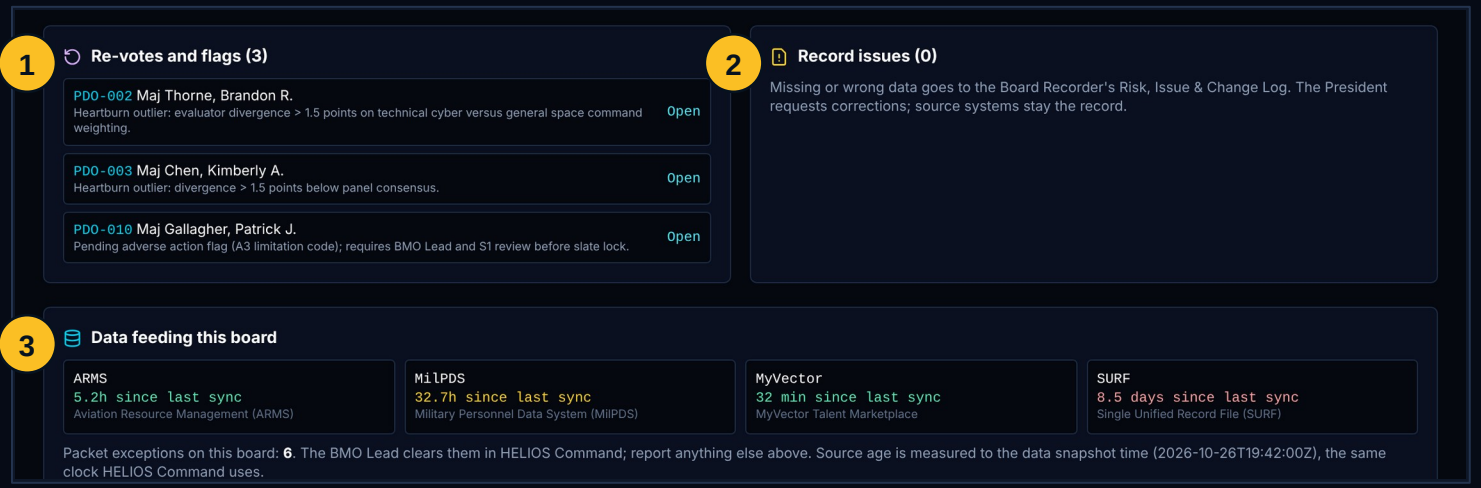
Rule Divergent = that voter's score is 1.5 or more points from the record's panel average. The 1.5 limit is an ORBIT design setting with no empirical basis yet.

Trust Calculated live

Panel Oversight: Re-votes, issues and data feeds (1 of 2)

The Board President's controls: what the panel sees from AI and when, panel progress, re-votes and record issues.

BMO Lead: View Read only.



1 Re-votes and flags

Shows Records flagged for discussion or called for a re-vote, with the reason. Open jumps to the record.

Data Candidate.flagged, Candidate.flagReason.

Rule A flag needs a reason. Only the President calls a re-vote (REVOTE_CALL).

Trust Calculated live + logged to DELTA

2 Record issues

Shows A form to report missing or wrong data on a record to the Board Recorder.

Data Risk, issue and change log.

Rule Needs 10 or more characters. Writes RECORD_ISSUE and adds an item to the Recorder's log. The source system stays the record.

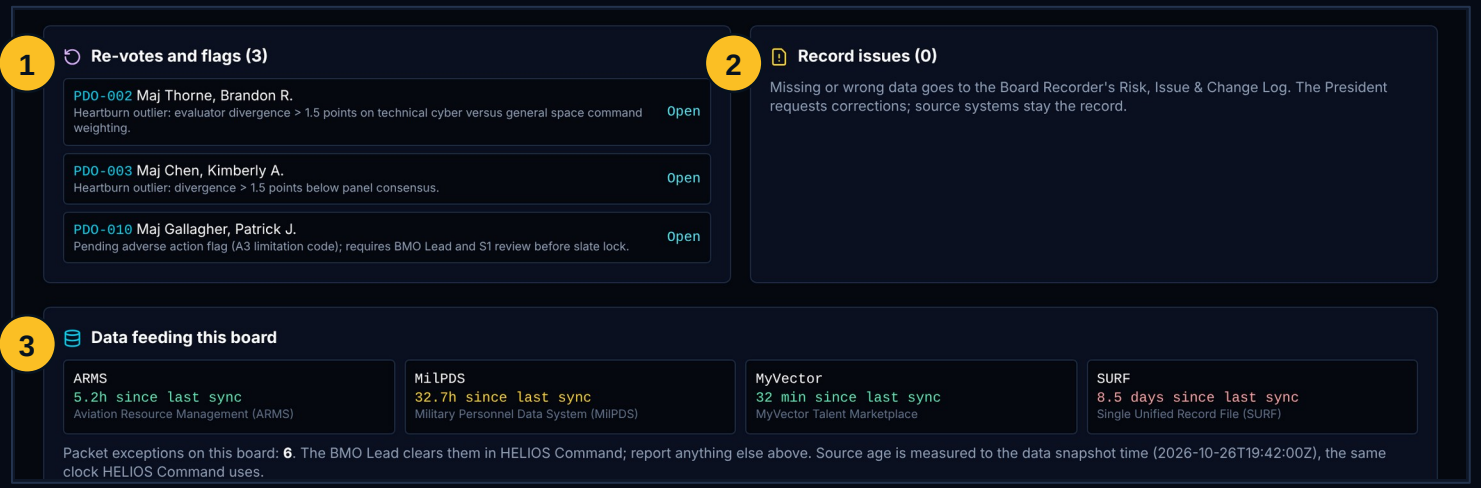
Trust Fixed rule + logged to DELTA

BMO Lead Form is read only.

Panel Oversight: Re-votes, issues and data feeds (2 of 2)

The Board President's controls: what the panel sees from AI and when, panel progress, re-votes and record issues.

BMO Lead: View Read only.



3 Data feeding this board

Shows Hours since each source last synced, and the count of packet exceptions on the board.

Data SourcePipeline.lastSync; Candidate.packetReadiness.exceptions.

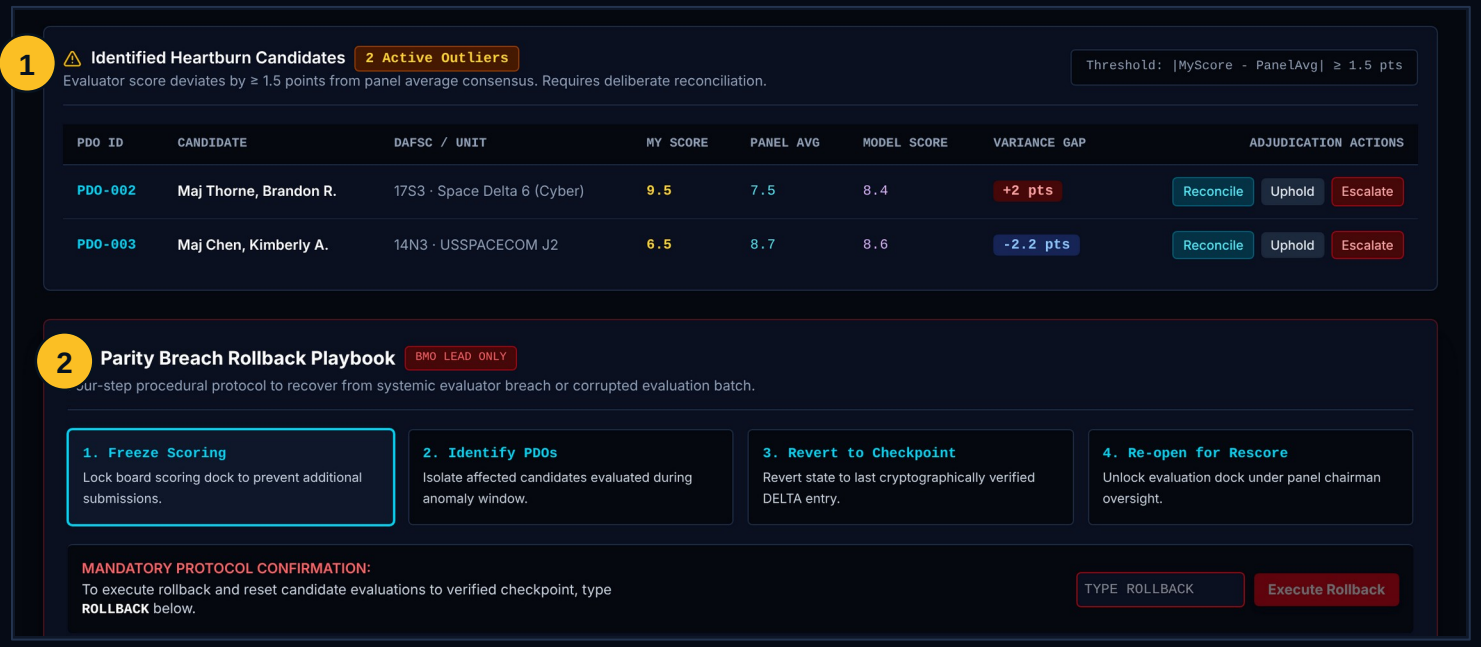
Rule Age = data snapshot time minus the stored last-sync time. Same function and limits as HELIOS Command.

Trust Calculated live. Matches HELIOS Command and Board Statistics.

Scoring Audit: Divergent scores and rollback (1 of 2)

Finds scores far from the panel average and records what a person decided to do about each.

BMO Lead: Uses Runs some actions here.



1 Divergent scores table

Shows Each record where a score sits 1.5 or more points from the panel average: the score, the panel average, the model score and the gap, with Reconcile, Uphold and Escalate.

Data Candidate.myScore, Candidate.scores, Candidate.mScore.

Rule Gap = score minus panel average. Listed when the gap is 1.5 or more either way. Every action needs a note and writes HEARTBURN_ADJUDICATE.

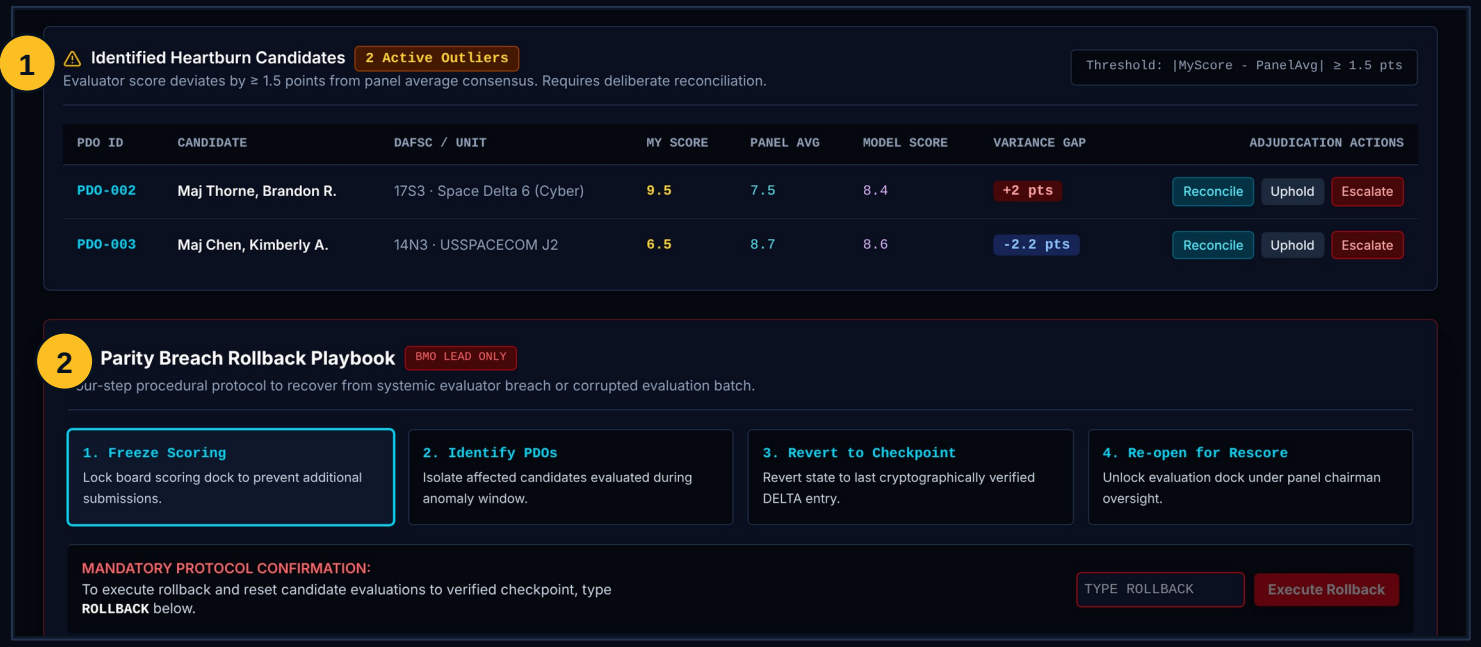
Trust Calculated live + logged to DELTA. In the demo "My Score" is the demo panelist seat's score for every role. The model score column is Pre-written stand-in.

BMO Lead May adjudicate with a note.

Scoring Audit: Divergent scores and rollback (2 of 2)

Finds scores far from the panel average and records what a person decided to do about each.

BMO Lead: Uses Runs some actions here.



2 Rollback playbook

Shows Four steps to recover from a bad scoring batch: freeze, identify, revert, reopen. Needs the word ROLLBACK typed to confirm.

Data Scores and the DELTA ledger.

Rule Writes ROLLBACK. Demo: one person (BMO Lead or President). Production decision: needs both the BMO Lead and the President.

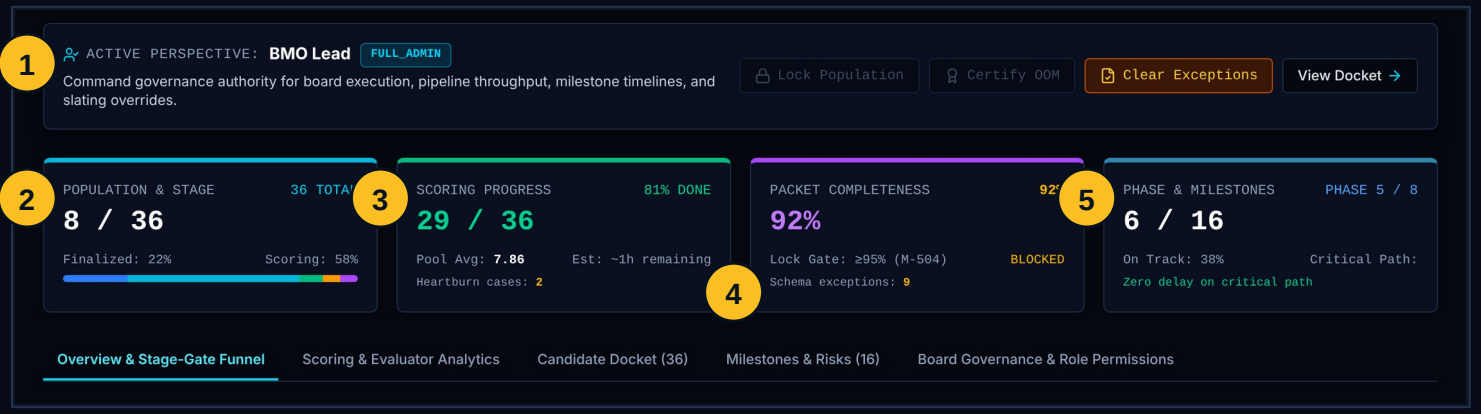
Trust Fixed rule + logged to DELTA

BMO Lead May run the rollback.

Board Statistics & Roles: Role banner and four cards (1 of 3)

One board's numbers, gates and role rights. This is where the population is locked and the Order of Merit is certified.

BMO Lead: Owns Runs this page's actions.



1 Role banner and gated buttons

Shows Your role, its access level, and the actions it may take: Lock Population, Certify OOM, Clear Exceptions, View Docket.

Data Role permission set; Board; packet completeness.

Rule Lock needs 95% packets. Certify writes BOARD_OOM_CERTIFY. Each button is enabled only for roles allowed to use it.

Trust Fixed rule + logged to DELTA. Enforced by the screen in the demo; by the data platform in production.

BMO Lead Locks, certifies and clears exceptions when the gates allow.

2 Population and stage

Shows Finalized records out of the pool, with the share at each stage.

Data Candidate.flow.stage.

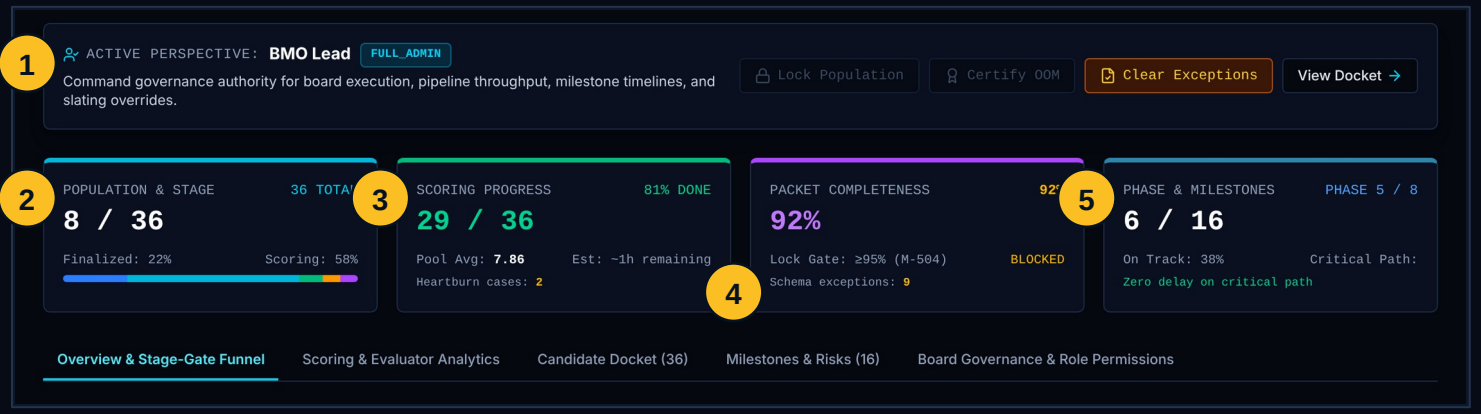
Rule Counts per stage divided by the pool.

Trust Calculated live

Board Statistics & Roles: Role banner and four cards (2 of 3)

One board's numbers, gates and role rights. This is where the population is locked and the Order of Merit is certified.

BMO Lead: Owns Runs this page's actions.



3 Scoring progress

Shows Records with at least one score, pool average, divergent cases and time remaining.

Data Candidate.scores.

Rule Scored = records with one or more scores. Pool average = mean of all scores.

Trust Calculated live for counts and average. Sample value for "time remaining" and pace.

4 Packet completeness

Shows Share of records with complete documents, against the 95% lock gate, and the exception count.

Data Candidate.packetReadiness.complete, exceptions.

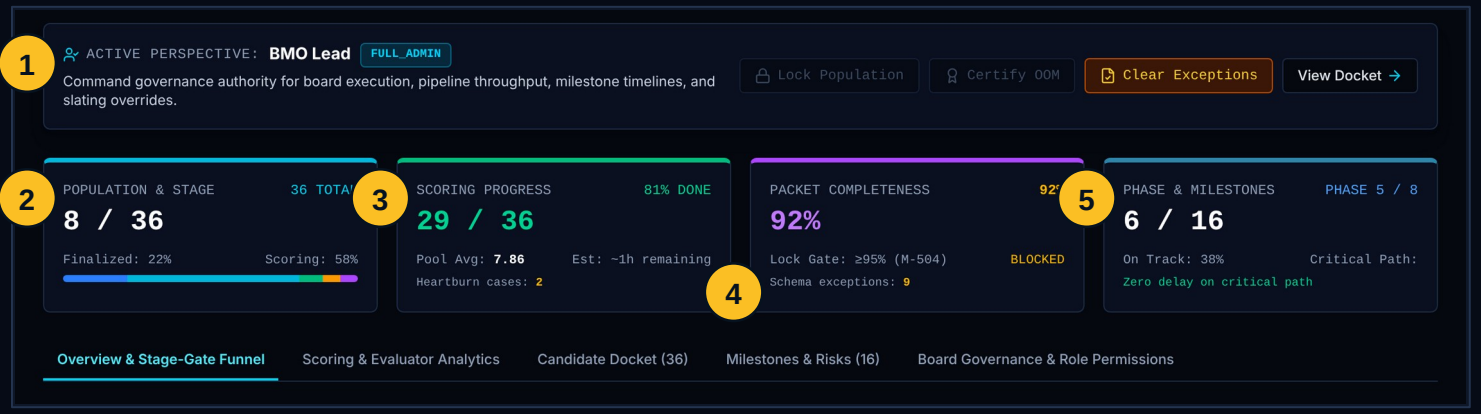
Rule Complete packets divided by records (33 of 36 = 92%). Same definition as M-504 on HELIOS Command.

Trust Calculated live

Board Statistics & Roles: Role banner and four cards (3 of 3)

One board's numbers, gates and role rights. This is where the population is locked and the Order of Merit is certified.

BMO Lead: Owns Runs this page's actions.



5 Phase and milestones

Shows Milestones done out of total, the current phase, and critical-path delay.

Data Milestone: phase, planned, actual, status.

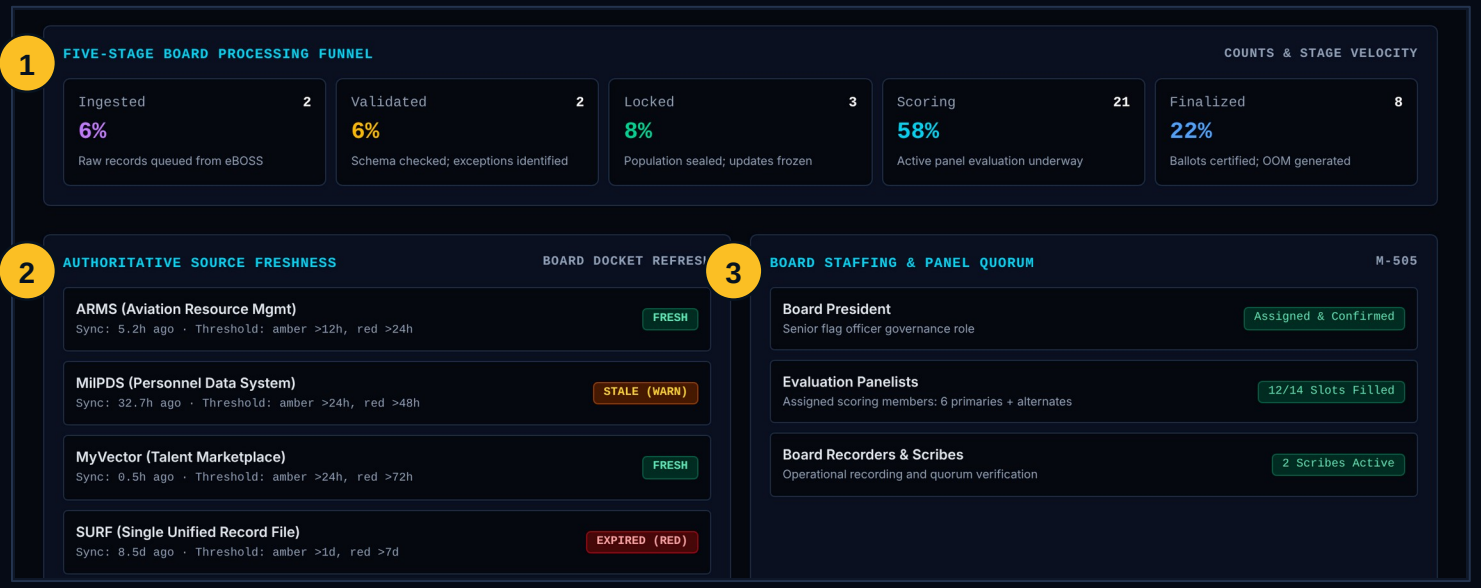
Rule Done count over total. Phase is inferred from stage counts. Delay is set from board status: 0, 2 or 6 days.

Trust Calculated live for counts. Fixed rule for phase and delay (not measured from dates).

Board Statistics & Roles: Tab 1: Overview and funnel

One board's numbers, gates and role rights. This is where the population is locked and the Order of Merit is certified.

BMO Lead: Owns Runs this page's actions.



1 Five-stage funnel

Shows Count and share of records at each of the five stages.

Data Candidate.flow.stage.

Rule Count per stage; share = count divided by pool.

Trust Calculated live

2 Source freshness for this board

Shows Age and status of ARMS, MilPDS, MyVector and SURF data.

Data SourcePipeline last sync time.

Rule Same function and limits as HELIOS Command.

Trust Calculated live (5.2h, 32.7h, 0.5h, 8.5 days). The sync times are fictional.

3 Staffing and quorum

Shows President, panelist seats filled, and scribes.

Data StaffSlot.

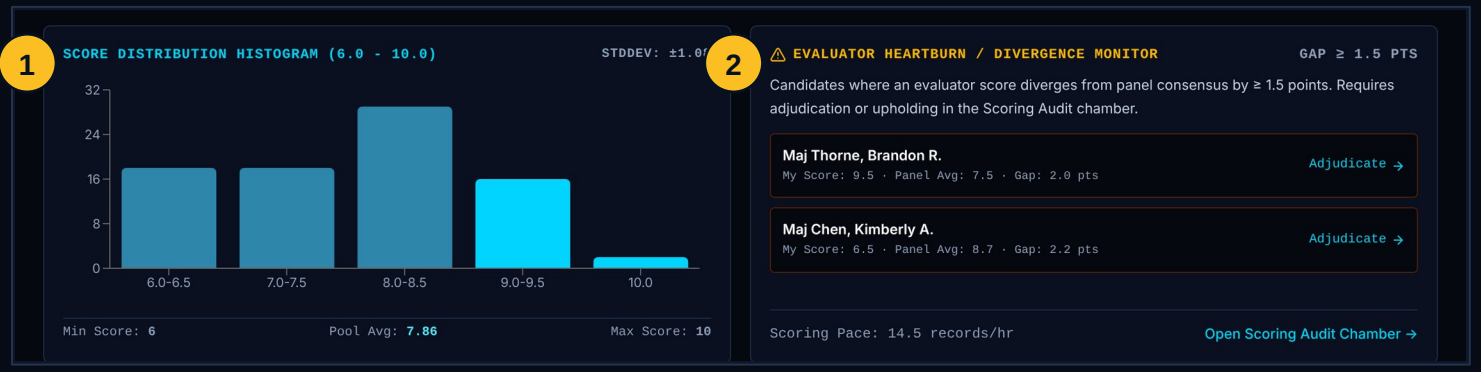
Rule Counts assigned seats by role.

Trust Calculated live, with fallbacks that always show the President and scribes as assigned

Board Statistics & Roles: Tab 2: Scoring analytics

One board's numbers, gates and role rights. This is where the population is locked and the Order of Merit is certified.

BMO Lead: Owns Runs this page's actions.



1 Score distribution

Shows How many scores fall in each half-point band from 6.0 to 10.0, with the standard deviation, minimum, pool average and maximum.

Data Every Candidate.scores value on the board.

Rule Population standard deviation: square root of the mean squared distance from the pool average.

Trust Calculated live

2 Divergence monitor

Shows Records where a score is 1.5 or more points from the panel average, with a link to adjudicate, and the scoring pace.

Data Candidate.myScore, Candidate.scores.

Rule Same 1.5-point rule as Scoring Audit.

Trust Calculated live for the list. Sample value for scoring pace (14.5 records per hour).

Board Statistics & Roles: Tab 3: Candidate docket

One board's numbers, gates and role rights. This is where the population is locked and the Order of Merit is certified.

BMO Lead: Owns Runs this page's actions.



1

CANDIDATE ID	OFFICER NAME	GRADE / DAFSC	LOOK STATUS	STAGE GATE	SCORES (POOL / MY)	PACKET CHECK	ACTION
PDO-001	Maj Vance, Tyler K.	MAJ · 13S3	1st Look	Scoring	9.0	Complete	Review
PDO-002	Maj Thorne, Brandon R.	MAJ · 17S3	1st Look	Scoring	7.5 (My: 9.5)	Complete	Review
PDO-003	Maj Chen, Kimberly A.	MAJ · 14N3	2nd Look	Scoring	8.7 (My: 6.5)	Complete	Review
PDO-004	Maj Morales, Javier E.	MAJ · 63A3	1st Look	Scoring	6.5	Complete	Review
PDO-005	Maj Sterling, Alyssa M.	MAJ · 61C3	1st Look	Scoring	6.5	Complete	Review
PDO-006	Maj Kowalski, Derek T.	MAJ · 13S3	1st Look	Scoring	6.5	Complete	Review
PDO-007	Capt O'Reilly, Sean P.	CAPT · 13S3	Below Zone	Scoring	8.0	1 Exception	Review
PDO-008	Maj Sterling-Fox, Diana	MAJ · 13S3	1st Look	Scoring	8.5	1 Exception	Review
PDO-009	Maj Al-Mansoor, Tariq	MAJ · 13S3	1st Look	Scoring	8.5	1 Exception	Review
PDO-010	Maj Gallagher, Patrick J.	MAJ · 13S3	2nd Look	Scoring	6.8	1 Exception	Review
PDO-011	Maj Washington, Maya L.	MAJ · 13S3	1st Look	Scoring	9.8	Complete	Review
PDO-012	Capt Lindqvist, Erik S.	CAPT · 17S3	1st Look	Validated	—	3 Exception	Review
PDO-013	Maj Ramirez, Carlos A.	MAJ · 13S3	1st Look	Ingested	—	Complete	Review
PDO-014	Capt Becker, Laura T.	MAJ · 61C3	1st Look	Validated	—	Complete	Review
PDO-015	Maj Harrison, Gregory K.	CAPT · 17S3	2nd Look	Ingested	—	2 Exception	Review

1 Candidate docket

Shows Every record on the board with its stage, packet status and scoring status. A row opens the record.

Data Candidate: flow.stage, packetReadiness, scores.

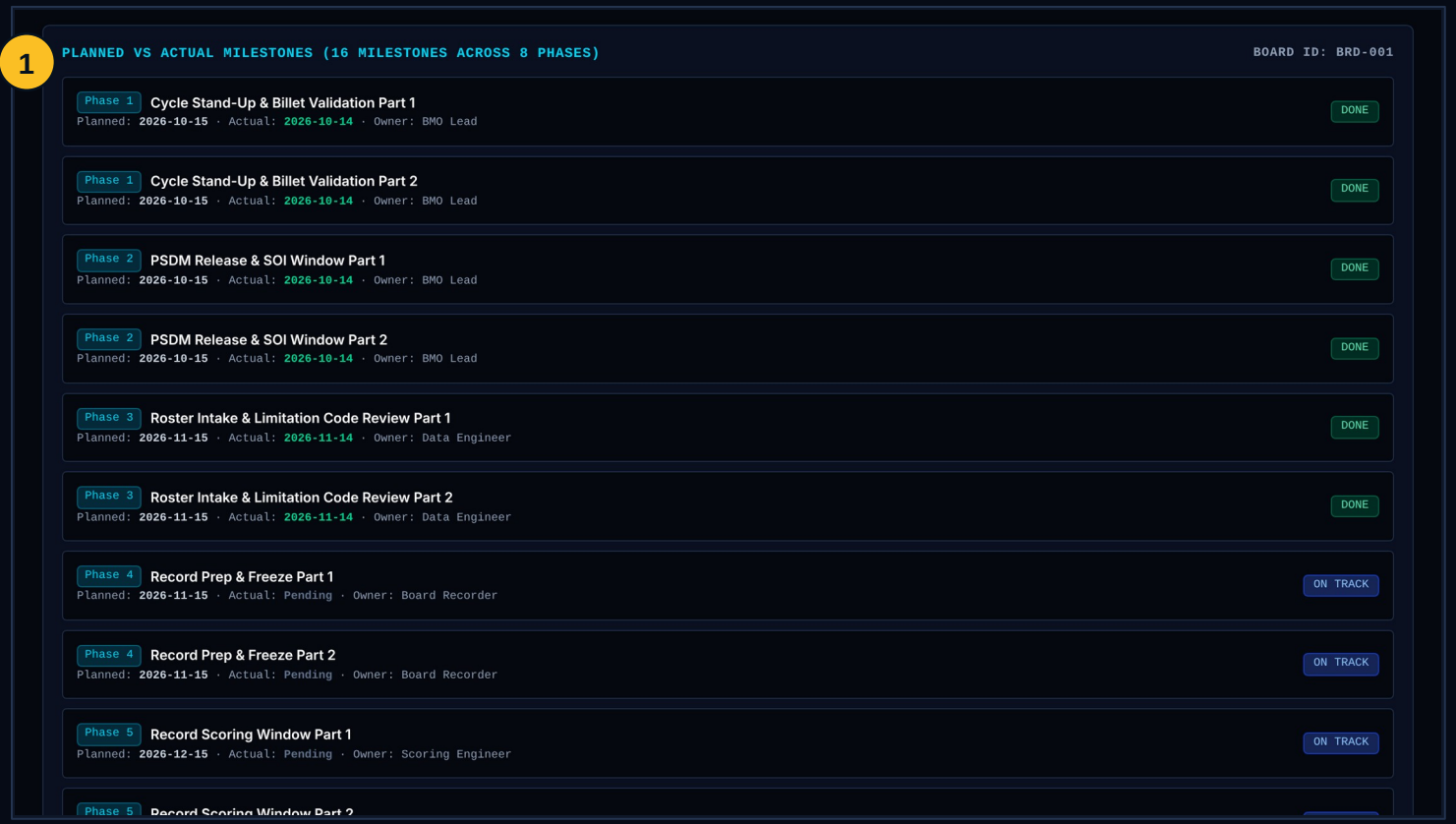
Rule Listed from the same records as the Candidate Queue.

Trust Calculated live

Board Statistics & Roles: Tab 4: Milestones

One board's numbers, gates and role rights. This is where the population is locked and the Order of Merit is certified.

BMO Lead: Owns Runs this page's actions.



1 Planned and actual milestones

Shows Sixteen milestones across eight phases: owner, planned date, actual date, status.

Data Milestone: phase, name, owner, planned, actual, status.

Rule Shown as stored. Status is a stored field, not worked out from the dates.

Trust Calculated live display

Board Statistics & Roles: Tab 4: Risks and issues

One board's numbers, gates and role rights. This is where the population is locked and the Order of Merit is certified.

BMO Lead: Owns Runs this page's actions.



1 Risks and issues

Shows Open risks, issues and changes for this board, with owner, suspense date and mitigation.

Data Risk: type, title, owner, suspense, escalate, status.

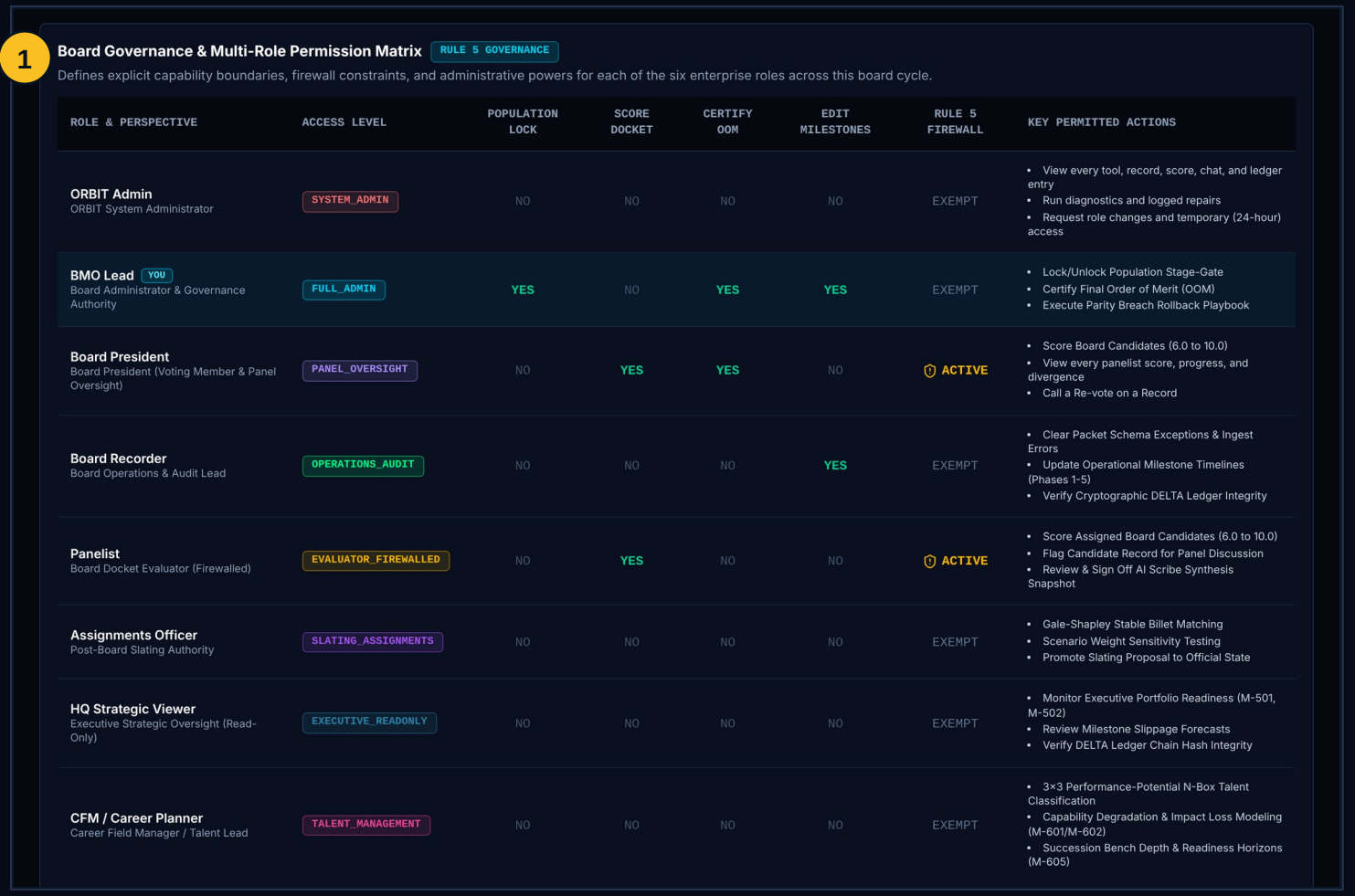
Rule Open items only. Record issues sent by the President land here.

Trust Calculated live

Board Statistics & Roles: Tab 5: Role permissions

One board's numbers, gates and role rights. This is where the population is locked and the Order of Merit is certified.

BMO Lead: Owns Runs this page's actions.



1 **Board Governance & Multi-Role Permission Matrix** RULE 5 GOVERNANCE

Defines explicit capability boundaries, firewall constraints, and administrative powers for each of the six enterprise roles across this board cycle.

ROLE & PERSPECTIVE	ACCESS LEVEL	POPULATION LOCK	SCORE DOCKET	CERTIFY OOM	EDIT MILESTONES	RULE 5 FIREWALL	KEY PERMITTED ACTIONS
ORBIT Admin ORBIT System Administrator	SYSTEM_ADMIN	NO	NO	NO	NO	EXEMPT	• View every tool, record, score, chat, and ledger entry • Run diagnostics and logged repairs • Request role changes and temporary (24-hour) access
BMO Lead YOU Board Administrator & Governance Authority	FULL_ADMIN	YES	NO	YES	YES	EXEMPT	• Lock/Unlock Population Stage-Gate • Certify Final Order of Merit (OOM) • Execute Parity Breach Rollback Playbook
Board President Board President (Voting Member & Panel Oversight)	PANEL_OVERSIGHT	NO	YES	YES	NO	ACTIVE	• Score Board Candidates (6.0 to 10.0) • View every panelist score, progress, and divergence • Call a Re-vote on a Record
Board Recorder Board Operations & Audit Lead	OPERATIONS_AUDIT	NO	NO	NO	YES	EXEMPT	• Clear Packet Schema Exceptions & Ingest Errors • Update Operational Milestone Timelines (Phases 1-5) • Verify Cryptographic DELTA Ledger Integrity
Panelist Board Docket Evaluator (Firewalled)	EVALUATOR_FIREWALLED	NO	YES	NO	NO	ACTIVE	• Score Assigned Board Candidates (6.0 to 10.0) • Flag Candidate Record for Panel Discussion • Review & Sign Off AI Scribe Synthesis Snapshot
Assignments Officer Post-Board Slating Authority	SLATING_ASSIGNMENTS	NO	NO	NO	NO	EXEMPT	• Gale-Shapley Stable Billet Matching • Scenario Weight Sensitivity Testing • Promote Slating Proposal to Official State
HQ Strategic Viewer Executive Strategic Oversight (Read-Only)	EXECUTIVE_READONLY	NO	NO	NO	NO	EXEMPT	• Monitor Executive Portfolio Readiness (M-501, M-502) • Review Milestone Slippage Forecasts • Verify DELTA Ledger Chain Hash Integrity
CFM / Career Planner Career Field Manager / Talent Lead	TALENT_MANAGEMENT	NO	NO	NO	NO	EXEMPT	• 3×3 Performance-Potential N-Box Talent Classification • Capability Degradation & Impact Loss Modeling (M-601/M-602) • Succession Bench Depth & Readiness Horizons (M-605)

1 Role permission matrix

Shows Each role and what it may do on a board: lock, certify, score, edit milestones, manage risks, see talent data, export.

Data Board role permission set.

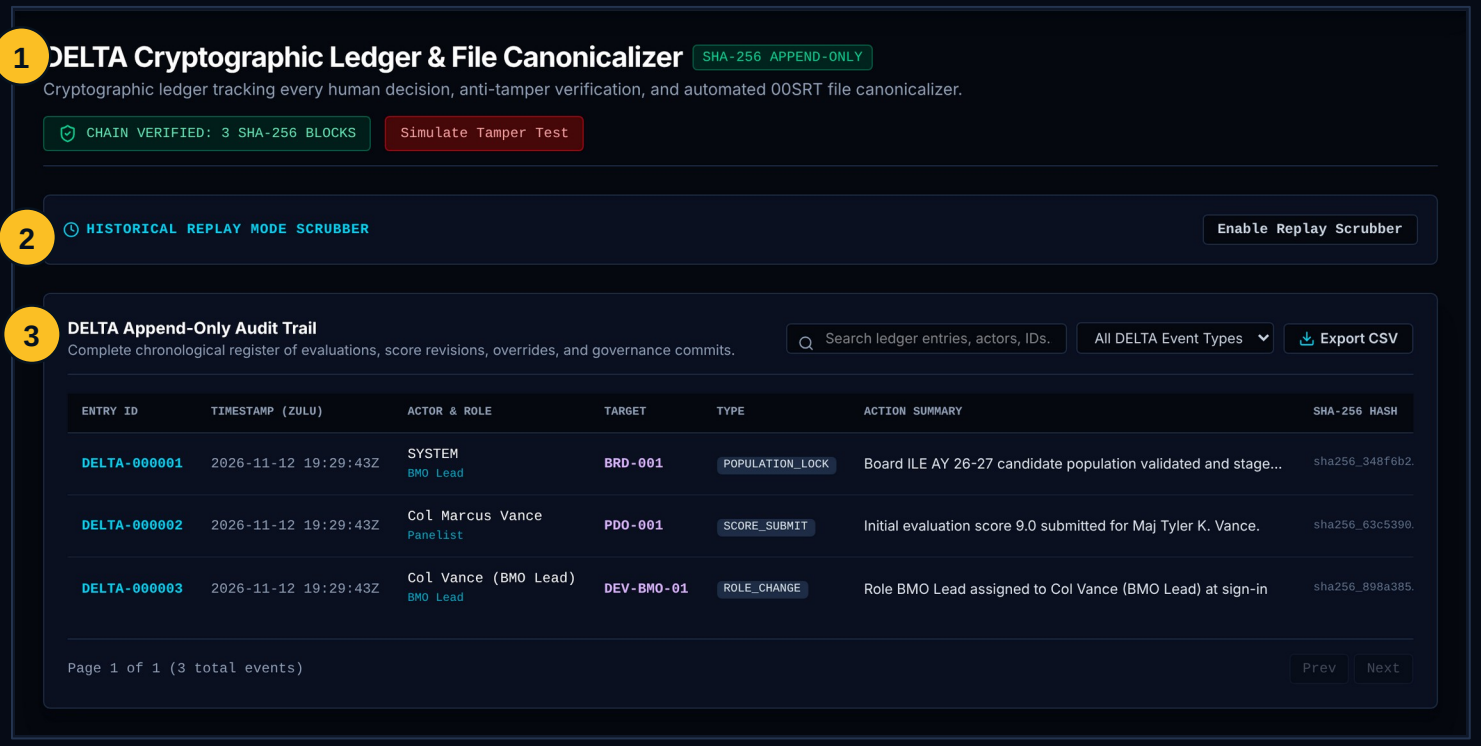
Rule Read-only reference. The live checks are in the permission table and in each action.

Trust Fixed rule

DELTA Ledger: Chain, replay and audit trail

The audit trail of every action, and the gate every incoming file passes through.

BMO Lead: View Read only.



1 Chain status and tamper test

Shows Whether the chain verifies, the number of entries, and a button that simulates tampering.

Data DeltaEntry.prevHash, DeltaEntry.hash.

Rule Each entry's SHA-256 hash covers its content and the previous entry's hash. Changing any entry breaks every hash after it.

Trust Calculated live

BMO Lead Runs the tamper test.

2 Replay scrubber

Shows Steps the board back to any earlier entry to see what was known at that point.

Data The ordered ledger.

Rule Read only. Replays entries in order up to the chosen point.

Trust Calculated live

3 Audit trail

Shows Every entry: ID, time (Zulu), actor and role, target, type, summary, hash. Search, filter by type, export CSV.

Data DeltaEntry.

Rule Entries are only added. A correction is a new entry. Refused actions are entries too (ACCESS_DENIED).

Trust logged to DELTA

BMO Lead Sees every entry.

DELTA Ledger: Late files after record freeze

The audit trail of every action, and the gate every incoming file passes through.

BMO Lead: View Read only.



1 Late file gate

Shows Files that arrived after the record freeze: canonical name, gate result, who received it, and the decision.

Data CanonicalFileRecord; FILE_ROUTED ledger entries marked late.

Rule A late file is renamed to the standard, always lands as WARN, and reaches the record only when the BMO Lead accepts it. The earlier version is kept. Both steps are logged.

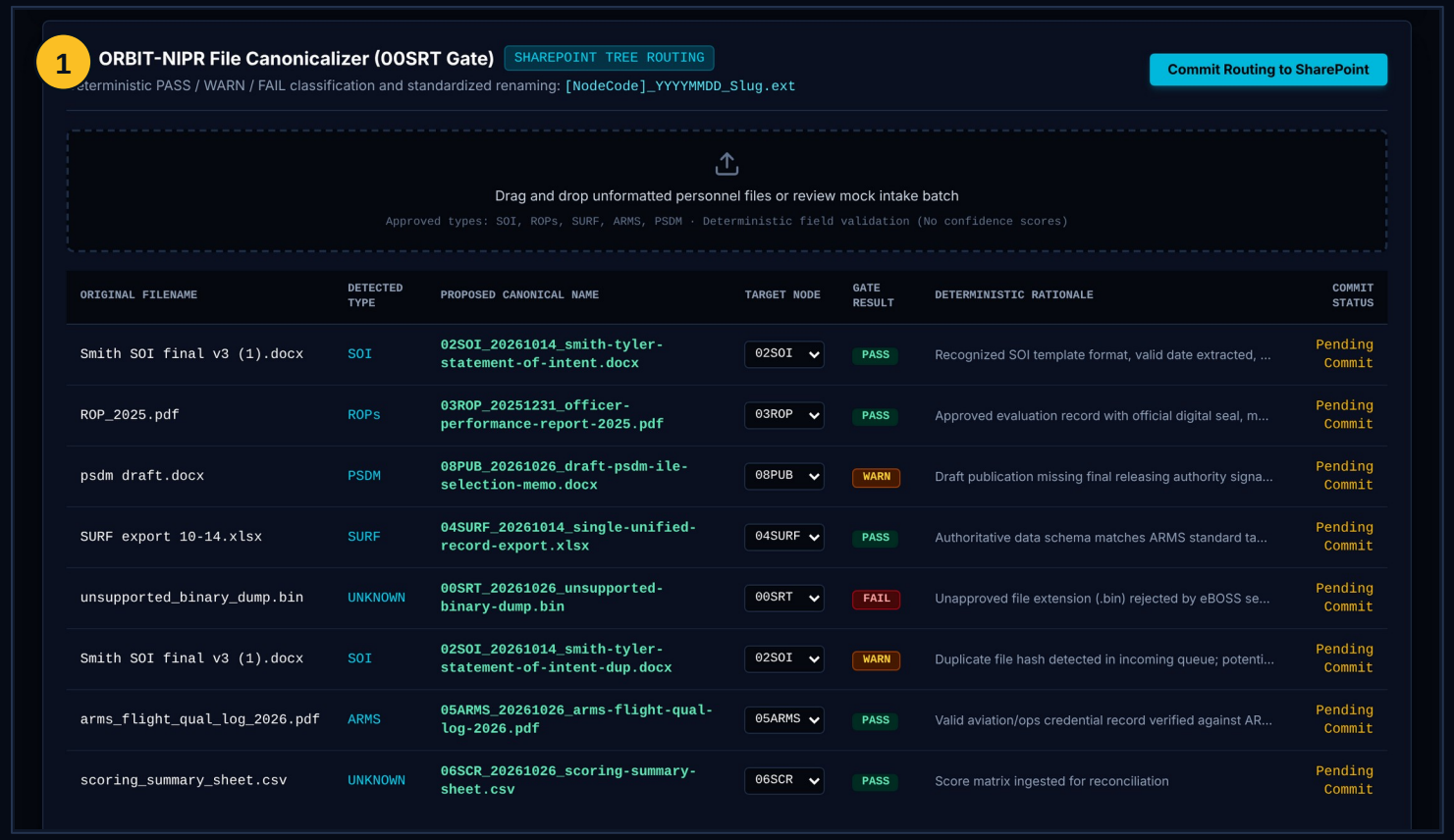
Trust Fixed rule + logged to DELTA

BMO Lead Receives and accepts late files.

DELTA Ledger: File canonicalizer (00SRT gate)

The audit trail of every action, and the gate every incoming file passes through.

BMO Lead: View Read only.



1 File canonicalizer

Shows Each incoming file: original name, detected type, proposed standard name, target folder, gate result and reason, commit status.

Data CanonicalFileRecord: originalName, detectedDocType, canonicalName, targetNode, checkResult, reason, committed.

Rule Standard name = folder code, date, short title, extension. PASS routes. WARN waits for a person. FAIL (bad type or duplicate) is rejected. No confidence scores: every result is a yes or no rule.

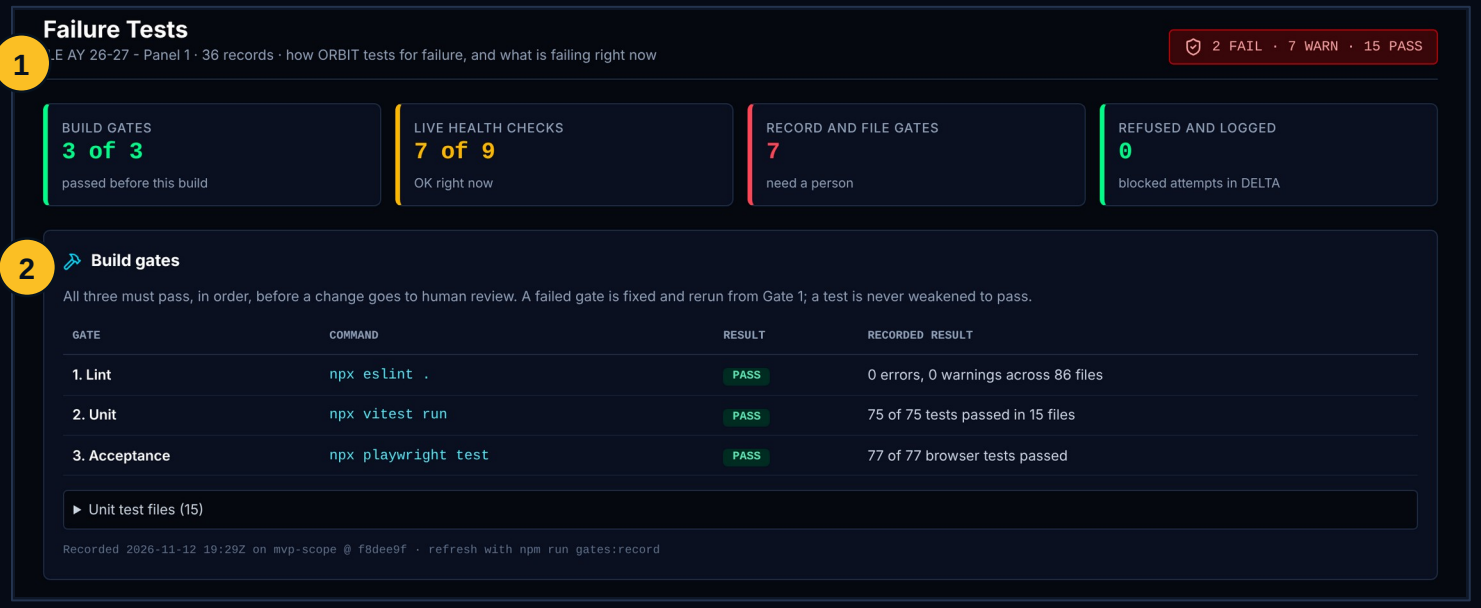
Trust Fixed rule + logged to DELTA (FILE_ROUTED). Method from the AI 101 Technical capstone.

BMO Lead Routes files and commits the routing.

Failure Tests: Summary and build gates

One screen that answers: how does ORBIT test for failure, and is anything failing right now?

BMO Lead: View Read only.



1 Summary and four tiles

Shows Counts of FAIL, WARN and PASS across every check on the page, and one tile per group.

Data The four sections below.

Rule Worst result in a group sets the tile color.

Trust Calculated live roll-up. It inherits Sample value values from the freshness rows.

2 Build gates

Shows The three checks every code change must pass before a person reviews it, with the recorded result.

Data Gate results file written by the test tools at build time.

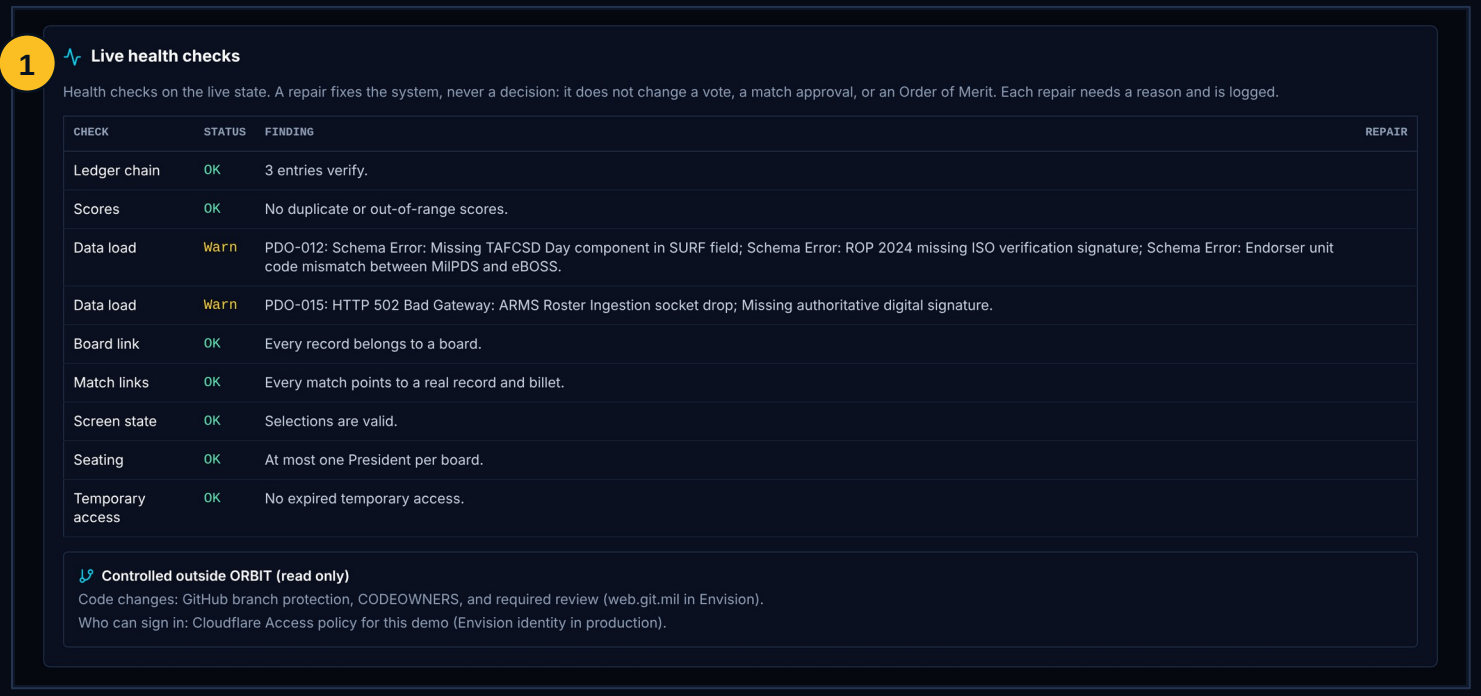
Rule On this build: lint 0 errors, unit 73 of 73, browser 77 of 77. A failed gate is fixed and rerun from Gate 1. A test is never weakened to pass.

Trust Recorded from real tool output for this build. Not typed.

Failure Tests: Live health checks

One screen that answers: how does ORBIT test for failure, and is anything failing right now?

BMO Lead: View Read only.



1

Live health checks

Health checks on the live state. A repair fixes the system, never a decision: it does not change a vote, a match approval, or an Order of Merit. Each repair needs a reason and is logged.

CHECK	STATUS	FINDING	REPAIR
Ledger chain	OK	3 entries verify.	
Scores	OK	No duplicate or out-of-range scores.	
Data load	Warn	PDO-012: Schema Error: Missing TAFCSD Day component in SURF field; Schema Error: ROP 2024 missing ISO verification signature; Schema Error: Endorser unit code mismatch between MilPDS and eBOSS.	
Data load	Warn	PDO-015: HTTP 502 Bad Gateway: ARMS Roster Ingestion socket drop; Missing authoritative digital signature.	
Board link	OK	Every record belongs to a board.	
Match links	OK	Every match points to a real record and billet.	
Screen state	OK	Selections are valid.	
Seating	OK	At most one President per board.	
Temporary access	OK	No expired temporary access.	

Controlled outside ORBIT (read only)

Code changes: GitHub branch protection, CODEOWNERS, and required review (web.git.mil in Envision).

Who can sign in: Cloudflare Access policy for this demo (Envision identity in production).

1 Live health checks

Shows Nine checks on the running data: ledger chain, duplicate or out-of-range scores, data load, board links, match links, screen state, seating, temporary access.

Data The whole app state.

Rule Each check is a yes or no test. A repair needs a reason of 20 or more characters and is logged. A repair never changes a vote, a match or the Order of Merit.

Trust Calculated live + logged to DELTA

BMO Lead Reads the findings.

Failure Tests: Record and file gates

One screen that answers: how does ORBIT test for failure, and is anything failing right now?

BMO Lead: View Read only.



1 **Record and file gates**

Read from the same metrics and file checks the board runs on. WARN waits for a named person; FAIL stays out until it is fixed.

GATE	TARGET	NOW	RESULT	FIX IN
Failed Ingests M-104	0	1 failed	FAIL	HELIOS Command →
Stuck Records (>24h) M-103	0	2 stuck	WARN	HELIOS Command →
MilPDS Freshness M-302	<= 24h	32.7h ago	WARN	HELIOS Command →
SURF Freshness M-304	<= 1 day	8.5 days ago	FAIL	HELIOS Command →
ARMS Freshness M-301	<= 12h	5.2h ago	PASS	HELIOS Command →
MyVector Freshness M-303	<= 24h	32 min ago	PASS	HELIOS Command →
Packet Completeness M-504	>= 95% to lock	92%	WARN	Board Statistics & Roles →
Staffing Status M-505	President assigned; Panelists 14/14	Pres: Assigned · Panel: 12/14	PASS	Board Statistics & Roles →
Incoming files	PASS routes; WARN waits for a person; FAIL is rejected	5 pass · 2 warn · 1 fail	WARN	DELTA Ledger →
Late files after record freeze	Reaches the record only after BMO Lead acceptance	0 waiting	PASS	DELTA Ledger →
Divergent scores	A score 1.5 or more points from the record average is reviewed	4 records	WARN	Scoring Audit →

1 Record and file gates

Shows Eleven gates with target, current value and result: failed and stuck records, four source ages, packet completeness, staffing, incoming files, late files, divergent scores.

Data The metric registry, file records and scores.

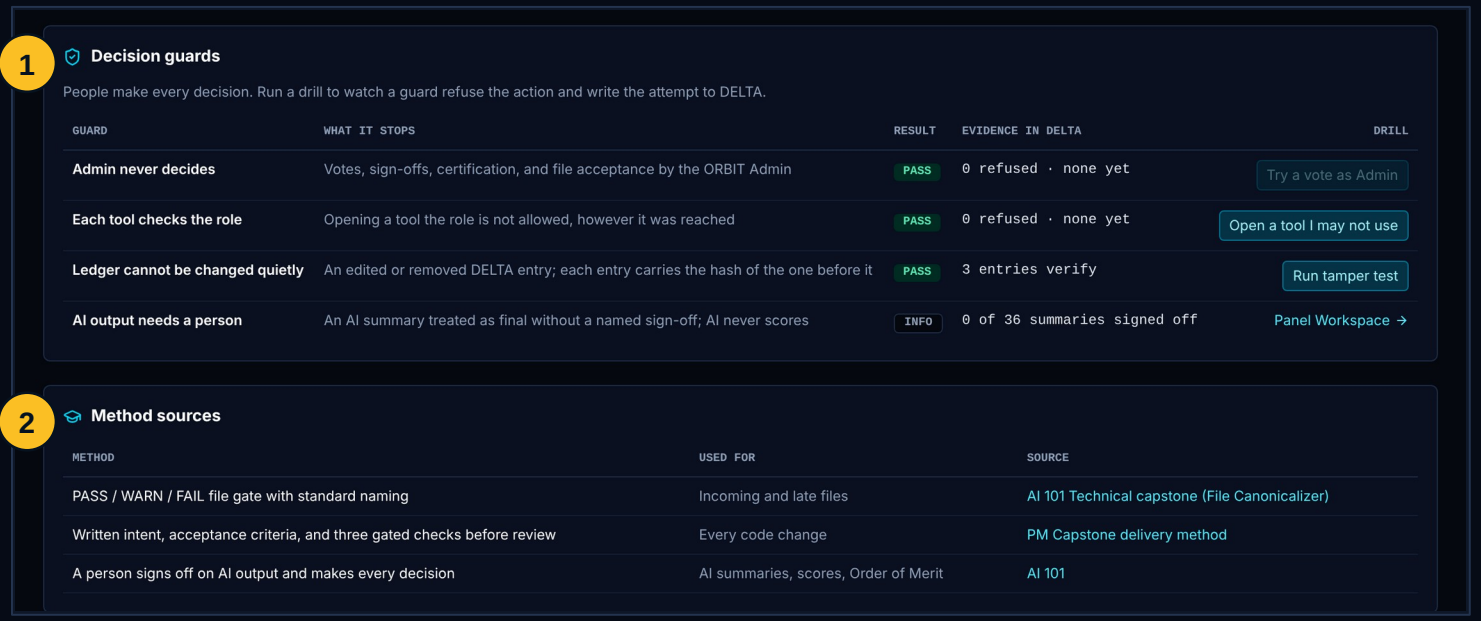
Rule Reads the same metric functions as HELIOS Command, so the numbers match that screen.

Trust Calculated live for failed, stuck, packets, staffing, files, late files and divergent scores. Sample value for the four source ages.

Failure Tests: Decision guards and method sources

One screen that answers: how does ORBIT test for failure, and is anything failing right now?

BMO Lead: View Read only.



1 Decision guards

People make every decision. Run a drill to watch a guard refuse the action and write the attempt to DELTA.

GUARD	WHAT IT STOPS	RESULT	EVIDENCE IN DELTA	DRILL
Admin never decides	Votes, sign-offs, certification, and file acceptance by the ORBIT Admin	PASS	0 refused · none yet	Try a vote as Admin
Each tool checks the role	Opening a tool the role is not allowed, however it was reached	PASS	0 refused · none yet	Open a tool I may not use
Ledger cannot be changed quietly	An edited or removed DELTA entry; each entry carries the hash of the one before it	PASS	3 entries verify	Run tamper test
AI output needs a person	An AI summary treated as final without a named sign-off; AI never scores	INFO	0 of 36 summaries signed off	Panel Workspace →

2 Method sources

METHOD	USED FOR	SOURCE
PASS / WARN / FAIL file gate with standard naming	Incoming and late files	AI 101 Technical capstone (File Canonicalizer)
Written intent, acceptance criteria, and three gated checks before review	Every code change	PM Capstone delivery method
A person signs off on AI output and makes every decision	AI summaries, scores, Order of Merit	AI 101

1 Decision guards

Shows Four guards with a drill button each: Admin never decides, each tool checks the role, the ledger cannot be changed quietly, AI output needs a person.

Data ACCESS_DENIED entries in DELTA; chain verification; summary sign-offs.

Rule A drill attempts the forbidden action. ORBIT refuses it and writes the attempt to DELTA, and the count on this row goes up.

Trust Calculated live + logged to DELTA

BMO Lead Runs the tamper test and "Open a tool I may not use".

2 Method sources

Shows Where each failure-testing method came from.

Data A short list kept in the code.

Rule Reference only.

Trust Typed list. Confirm each source before briefing.

What to say if asked "is that number real?" (1 of 3)

49 of 69 widgets this role sees are fully calculated, rule-based or logged. These 20 are not, in whole or in part.

Screen	Widget	Status	The plain answer
HELIOS Command	**KPI strip (M-101 to M-105)**	**Mixed**	Calculated live for the processed count, stuck and failed. Sample value for "+12% vs prior", "4.2 min" and "Yes (14d remaining)".
HELIOS Command	**Throughput trend (M-106)**	**Sample**	Sample value. The seven points and the +1.45 slope are fixed in the demo.
HELIOS Command	**Readiness gauges (M-501, M-502, M-504)**	**Mixed**	Calculated live for packet completeness. Sample value for readiness 96% and at-risk 4%.
Candidate Queue	**Board header and counters**	**Mixed**	Calculated live for counts, Board Avg and My Avg. Sample value for "Panel": it is Board Avg plus 0.1, and My Avg shows a placeholder until you score.
Candidate Queue	**Record table**	**Stand-in**	Calculated live for packet status and scores. Pre-written stand-in for the AI fit value: a number typed per record, not produced by a model.
Panel Workspace	**Statement of Intent**	**Stand-in**	Calculated live display of Pre-written stand-in fictional text
Panel Workspace	**Evaluations and stratifications**	**Stand-in**	Calculated live display; trend is Pre-written stand-in
Panel Workspace	**Competency clusters and skill vector**	**Stand-in**	Pre-written stand-in. The eight values are typed per record. No method yet derives them from a real record.
Panel Workspace	**AI fit, model score and feature drivers**	**Stand-in**	Pre-written stand-in. All four are typed per record. No model runs in the demo, so these numbers have no validity evidence.

What to say if asked "is that number real?" (2 of 3)

49 of 69 widgets this role sees are fully calculated, rule-based or logged. These 20 are not, in whole or in part.

Screen	Widget	Status	The plain answer
Panel Workspace	**Statement of Intent, full**	**Stand-in**	Calculated live display of Pre-written stand-in fictional text
Panel Workspace	**Endorser context (M-404, M-405)**	**Stand-in**	Pre-written stand-in. Values are typed per endorser. A real version needs endorsement history from the source system.
Panel Workspace	**Similar records (top 5)**	**Stand-in**	Calculated live math on Pre-written stand-in vectors. Because all values are positive, scores bunch near 98 to 100% and separate records poorly.
Panel Workspace	**Possible billet fits (top 3)**	**Stand-in**	Calculated live math on Pre-written stand-in vectors
Panel Workspace	**AI summary (frozen copy)**	**Stand-in**	Pre-written stand-in. The text is pre-written per record. No language model runs in the demo.
Scoring Audit	**Divergent scores table**	**Stand-in**	Calculated live + logged to DELTA. In the demo "My Score" is the demo panelist seat's score for every role. The model score column is Pre-written stand-in.
Board Statistics & Roles	**Scoring progress**	**Mixed**	Calculated live for counts and average. Sample value for "time remaining" and pace.
Board Statistics & Roles	**Divergence monitor**	**Mixed**	Calculated live for the list. Sample value for scoring pace (14.5 records per hour).
Failure Tests	**Summary and four tiles**	**Mixed**	Calculated live roll-up. It inherits Sample value values from the freshness rows.

What to say if asked "is that number real?" (3 of 3)

49 of 69 widgets this role sees are fully calculated, rule-based or logged. These 20 are not, in whole or in part.

Screen	Widget	Status	The plain answer
Failure Tests	**Record and file gates**	**Mixed**	Calculated live for failed, stuck, packets, staffing, files, late files and divergent scores. Sample value for the four source ages.
Failure Tests	**Method sources**	**Sample**	Typed list. Confirm each source before briefing.